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MILITARY SPECIFICATION

INTERROGATOR SET AN/APX-76A(V)

This specification has been approved by the
Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall provide an Air-by-Air IFF Interrogator Responsor which, when used in conjunction with appropriate associated equipment, is capable of transmitting coded signals to suitable transponder equipped aircraft, surface ships, and ground stations. It shall receive the coded replies from these transponders, process the reply information, and provide video outputs for display on associated radar indicators.

*1.2 Classification - The equipments covered by this specification shall consist of the following items:

Item	Type Designation	Applicable Paragraph
Control, Interrogator Set	C-7383/APX-76(V)	3.5.1
Control, Interrogator Set	C-7994/APX-76A(V)	3.5.1
Control, Interrogator Set	C-7959/APX-76A(V)	3.5.1
Receiver-Transmitter, Radio	RT-868A/APX-76(V)	3.5.2
Switch-Amplifier	SA-1568A/APX-76(V)	3.5.3
Synchronizer, Electrical	SN-416A/APX-76(V)	3.5.4
Receiver-Transmitter, Radio	RT-988/A	3.5.5
Base, Shock Mount, Electrical Equipment	MT-3923/APX-76(V)	3.5.6
Base, Shock Mount, Electrical Equipment	MT-3924/APX-76(V)	3.5.7
Base, Shock Mount, Electrical Equipment	MT-4024/APX-76A(V)	3.5.8

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1.2.1 Equipment Configurations - The AN/APX-76A(V) is capable of various configurations, depending on the control used:

Configuration	Equipment Items
(V)1	C-7383/APX-76(V), RT-868A/APX-76(V), SA-1568A/APX-76(V), SN-416A/APX-76(V)
(V)2	C-7959/APX-76A(V), RT-868A/APX-76(V), SN-416A/APX-76(V)
(V)3	C-7994/APX-76A(V), RT-868A/APX-76(V), SA-1568A/APX-76(V), SN-416A/APX-76(V)

1.3 Associated Equipment - The interrogator set is intended to operate with the following associated equipment:

Item	Common Name Used in this Specification	Appl. Para.
(1) Interrogator Computer KIR-1A/TSEC	Interrogator Computer	6.11.1
(2) Antenna AS-1900/APQ-72 (Radar Antenna Modification Kit) or equivalent	Interrogator Antenna	6.11.2
(3) Radar Modification Kit	---	6.11.3
(4) Radome Modification Kit	---	6.11.4

* The information contained in paragraphs 6.11.2, 6.11.3 and 6.11.4 is applicable only to the F-4B system. Detailed specifications for these requirements should be obtained from cognizant radar and antenna personnel.

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*2. APPLICABLE DOCUMENTS

*2.1 The following documents, of issue in effect on date of invitation for bids, unless otherwise specified, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-172	Cases, Bases, Mounting; and Mounts, Vibration
MIL-W-5088	Wiring; Aircraft, Installation of
MIL-E-5400 P	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-C-6781	Control Panel; Aircraft Equipment Rack or Console Mounted
MIL-P-7788	Plate, Plastic, Lighting
MIL-M-7793	Meter, Time Totalizing
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures, Preproduction and Inspection for Aircraft Electronic Equipment, format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment

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MIL-T-23103	Thermal Performance Evaluation, Airborne Electronic Equipment; General Requirement for
MIL-S-23603	System Readiness/Maintainability; Avionic Systems Design, General Specification for

Naval Air Systems Command

AR-5	Microelectronic Devices Used in Avionic Equipment, Procedures for Selection and Approval of
AR-34	Failure Classification for Reliability Testing, General Requirements for

DOD-AIMS System Program Office

DOD-AIMS 64-900A	Mode 4 Input/Output data, KIR/KIT-1A/TSEC.
DOD-AIMS 65-1000	Technical Standard for the ATCRBS/IFF Mark XII systems
DOD-AIMS 69-253	Performance/Design and Qualification Requirements, Interrogator Set AN/APX-83(V)

F-111 Aircraft

Exhibit SENX-LM-66-1 dated 5 Aug 66	Panels-Integrally Illuminated, White F-111 Aircraft

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STANDARDS

Military

MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-785	Requirements for Reliability Program (For Systems and Equipments)
MIL-STD-794	Parts and Equipment, Procedure for Packaging and Packing of
MS-17322	Meter, Time Totalizing, 115 Volt, 400-cycle, Rev. B

DRAWINGS

DOD/AIMS Drawing No. X-66B-1675	Installation Information Wiring Diagram, Interrogator Set AN/APX-76

OTHER

Naval Air Systems Command

EI-516A	Avionics Installation Instructions for Interrogator Set AN/APX-76A(V) and Receiver-Transmitter RT-988/A, dated March 31, 1972.
ET-516A	Bench, Preflight & Flight Test Instructions for Interrogator Set AN/APX-76A(V) and Receiver-Transmitter RT-988/A, dated March 31, 1972.

DOD-AIMS System Program Office

DOD-AIMS Doc. No. 134	Bench, Preflight and Flight Test Procedures for Interrogator Set AN/APX-76A, dated 10 June 1969.

NSA

NSA CSEEB-17A	Communications Security Equipment Engineering Bulletin, TSEC/K1-1A, Security Equipment for IFF Mark XII (U) dated March 1967 (CONFIDENTIAL Document).

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*2.2 Availability of Documents - When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Publications and Forms Center, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120, except for DOD/AIMS documents which are available from Electronic Systems Division, Air Force Systems Command, USAF L.G. Hanscom Field, Bedford, Mass. 01730, Code DCT. F-111 documents are available from ASD, Wright Patterson AFB, Ohio, Attn: SENX.

3. REQUIREMENTS

3.1 Preproduction - This specification makes provision for preproduction testing.

3.2 Materials, Parts and Processes - In the selection of materials, parts and processes, fulfillment of major design objectives shall be the prime consideration. In so doing, the following factors shall govern:

*3.2.1 Solid State Processes and Microelectronic Modular Assemblies - The electronic portion of the equipment shall be constructed using solid state processes and microelectronic modular assemblies to the greatest extent possible. Production type microelectronic modular assemblies produced by industrial producers of demonstrated competence shall be used. In the event production type microelectronic modular assemblies are not available, the contractor shall first consider establishing (either at the contractor's facilities or any other industrial producer's facilities) a production line for the microelectronics modular assemblies required. He may utilize conventional electron devices provided the nonsuitability of microelectronic modular assemblies is clearly established by the contractor and approved by NAVAIR. In the design and construction of microelectronic functional dvices, Specification AR-5 shall apply.

3.2.1.1 Microelectronic Techniques - Microelectronic techniques, defined as follows, shall be selected in the order of precedence shown:

(1) Integrated circuits - An integrated circuit is an assembly in which all of the functional elements, active and passive, are formed in place by basically the same process techniques.

(a) Semiconductor - The functional elements are formed by semiconductor techniques, i.e., diffusion, alloying doping, conductor patterns are deposited.

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(b) Thin film - All elements formed by deposition processes.

(2) Hybrid circuit - a hybrid circuit is an assembly in which the passive elements are formed by thin film techniques and the active elements are discrete devices.

(3) Discrete component circuit - A discrete component circuit is an assembly which is achieved by the interconnection of microelements, pellet components, or microcomponents.

3.2.2 Other Electronic Devices - Subject to the requirements of 3.2.1 herein giving primary consideration to microelectronic modular assemblies, electron devices other than microelectronic modular assemblies (i.e., semiconductor diodes and transistors not used as an integral part of a microelectronic modular assembly or electron tubes) shall be chosen and applied as outlined in Specification MIL-E-5400.

3.2.3 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials other than microelectronic devices shall be obtained as outlined in MIL-E-5400. Microelectronic devices shall be approved as outlined in AR-5.

3.3 Design and Construction - The equipment shall fulfill all applicable design, construction and workmanship requirements of Specification MIL-E-5400, unless otherwise specified herein.

3.3.1 Design Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the existing standards are objectives in the design of the equipment specified herein.

3.3.2 Total Weight - The total weight of the equipment components excluding cables shall not exceed 37 pounds.

3.3.3 Reliability

*3.3.3.1 Operational Stability - The equipment shall operate with satisfactory performance continuously or intermittenly, for a period of at least 225 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.3.3.2 Operating Life - The equipment shall have a total operating life of 10,000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

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*3.3.3.3 Equipment Reliability in Mean Time Between Failures (MTBF)- The interrogator set shall have 225 hours of mean (operating) time between failures (θ_o of MIL-STD-781) when tested and accepted as outlined under the requirements of 4.4.3.

*3.3.3.3.1 Reliability of Individual Items - Items for system calculations of 225 hours MTBF.

C-7383/APX-76(V)	22,500 hours
RT-868A/APX-76(V)	340 hours
SA-1568A/APX-76(V)	1,140 hours
SN-416A/APX-76(V)	1,770 hours

*3.3.3.4 Elapsed Time Indicator - An elapsed time indicator, meeting the requirements of MS17322-6A, in accordance with the requirements of MIL-M-7793, shall be included with each receiver-transmitter.

3.3.4 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400.

*3.3.4.1 Interconnecting Cabling - The equipment shall be capable of operating satisfactorily using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet the interference control requirements, and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment.

3.3.5 Control Panels - All rack or console mounted control panels shall conform to the applicable requirements of Specification MIL-C-6781. The configuration of all control panels must be approved by the Naval Air Systems Command prior to preproduction testing.

*3.3.6 Interchangeability - The equipment shall meet the interchangeability requirements of specification MIL-E-5400.

3.3.7 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181.

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3.3.7.1 Electronic Counter Countermeasures - The design of this equipment shall minimize vulnerability to interference, both to enemy jamming and friendly interference. Careful consideration shall be given to all known methods of receiving through enemy jamming and the method(s) shall be chosen which will most expeditiously achieve the desired immunity. The most effective anti-jam design features that can be obtained compatible with other performance requirements shall be incorporated.

3.3.8 Modules, Maintenance - The electronic portions of the equipment shall be divided into maintenance modules (see Specification MIL-S-23603). Maintenance modules shall normally be considered repairable.

3.3.8.1 Test Points - Modules and circuit card assemblies of the units shall contain adequate test points for troubleshooting, detailed performance checks and adjustments. Major test points shall be designed to facilitate fault isolation to the defective circuit card assembly or module. These test points shall be located to permit access during troubleshooting without displacement or removal of wiring or parts. Minor test points shall be provided to facilitate fault isolation to the defective stage or stages, when space permits. Potentials impressed on test points shall not exceed 1000 volts, rms. Where higher voltages are to be monitored, these voltages shall be stepped down accordingly. Test points having 350 to 1000 volts rms impressed thereon shall have barriers or guards provided to prevent accidental contact with such voltages and a warning plate shall be provided and shall be located in close proximity to the test points.

3.3.8.2 Fault Isolation - The units shall contain built-in fault isolation circuitry to enable in-position functional checking. The test circuitry shall be as complete as practicable to allow GO/NO-GO testing of the equipment and fault isolation to the defective replaceable assembly or subassemblies.

3.3.8.3 Fault Indicators - Fault indicators shall be provided on the front panels of the units to show their operational status. In addition to the front panel fault indicators, provisions shall be made to enable a remote fault indicator whenever any one or more of the front-panel fault indicators are enabled. The front-panel fault indicators of the receiver-transmitter, switch amplifier, and synchronizer units shall be of the type which will retain its last indication after power is removed. The receiver-transmitter shall contain 3 fault indicators, to indicate a transmitter fault, a receiver fault, and a video fault. The switch amplifier and synchronizer units shall each contain a fault indicator.

3.3.8.4 Maintenance and Repair Cycles - Scheduled (preventive) maintenance requirements shall be specified by the contractor in accordance with Specification MIL-E-5400 and forwarded to the procuring activity for approval.

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3.3.8.5 Reversibility - The units shall incorporate features such that it is mechanically or electrically impossible to install incorrectly its plug-in modules, circuit card assemblies, sub-assemblies and other items which when improperly installed, will result in damage to this or associated equipments.

3.3.9 Mounting - The interrogator control shall be mounted without the use of vibration isolators. The other components of the interrogator set shall be secured to vibration isolated mounts or a vibration isolated shelf.

*3.3.10 Nomenclature and Nameplates - Nomenclature and serial number assignment, nameplate approval and equipment identification shall be in accordance with Specification MIL-N-18307.

3.3.11 Standard Conditions - The following conditions shall be used as a basis to establish normal performance requirements and for making laboratory bench tests on the equipment.

Temperature	Room ambient 25°C ±5°C
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input Voltage	115 ±1.0 vac 380 to 420 Hz and 27.5 ±0.5 vdc

*3.3.12 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified herein and in specification MIL-E-5400 for Class 2 and Class 2X equipments. Classification 2X applies only to the receiver-transmitter.

3.3.12.1 Vibration - The equipment shall meet the vibration requirements of Curves I and II of MIL-E-5400, except that the receiver-transmitter shall meet requirements of Curves II and IV.

*3.3.12.2 Temperature - There shall be continuous and intermittent operation over the range defined in Specification MIL-E-5400 for Class 2 and Class 2X equipments.

* 3.3.12.2.1 Cooling - Provisions for cooling this equipment shall be employed to maintain components within their maximum temperature under all operating conditions in accordance with Specification MIL-E-5400. The cooling design shall be verified in accordance with MIL-T-23103.

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*3.3.12.2.2 Forced Air Cooling - The receiver-transmitter, when not installed on the MT-4024/APX-76A(V) receiver-transmitter mount, shall be supplied with forced air in accordance with figure 1 for operation at a one percent transmitting rf duty cycle. Figure 2 supplies back pressure data on the receiver transmitter operating at a specified flow rate. Receiver-transmitters installed on the receiver-transmitter mount can operate in an environment as shown in figure 3 for a 0.6 percent transmitting rf duty cycle. Verification of cooling design shall be in accordance with MIL-T-23103.

*3.3.12.3 Altitude - The equipment shall meet the altitude requirements for Class 2 and Class 2X equipments, except the maximum operating altitude for the receiver transmitter installed on the MT-4024/APX-76A(V) receiver-transmitter mount is 55,000 feet as indicated in figure 3. All equipments can operate at altitudes up to 70,000 ft, provided the receiver-transmitter is supplied with sufficient forced air cooling.

3.3.13 Warm-Up-Time - The maximum warm-up-time for the equipment shall not exceed 3 minutes.

*3.3.14 Power Sources - The interrogator set shall operate from power sources as defined in MIL-STD-704 for Category "B" equipment. The power consumption from each of the listed sources shall not be more than the following:

Source	Power
115 volts 400 Hz Single Phase Secondary Buss, Category "B"	265 volt-amperes
28 volts dc Secondary Buss Category "B"	31 watts

3.3.14.1 Degraded Performance - Degraded performance will be permitted for voltage transients not exceeding 0.5 second during normal electric system operation. With the application of an overvoltage transient, operation shall return to normal within a length of time not exceeding twice the transient time, with no resulting damage to the equipment.

3.3.14.2 AC Frequency - In general, the equipment shall be designed for normal operation over the frequency range of 380 to 420 Hz with operation at lower and higher frequencies of 320 to 480 for limited periods of time as specified in MIL-STD-704. Operation at these lower and higher frequencies may reduce performance but shall result in no damage to the equipment and the equipment shall resume normal operation when the frequency returns within the limits of 380 to 420 Hz.

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3.3.15 Overload Protection - Overload protection shall be provided in the equipment. All parts and circuits of the equipment which are likely to carry an overload due to any failures or poor adjustments shall be proportioned to withstand such overload without permanent damage to the equipment or shall have suitable protective devices. The use of fuses and other protective devices shall be held to a minimum.

3.3.15.1 Undervoltage Protection - The equipment shall not be damaged by voltage below the minimum specified herein and shall automatically resume normal operation when the voltage returns within limits.

3.4 Performance

3.4.1 General Performance Requirements - The interrogator set shall provide a means of identification of suitably equipped friendly aircraft. The interrogator set with the associated equipment shall generate and radiate coded Mode 1, 2, 3/A or 4 interrogations upon command of the operator. The interrogator set shall receive and process, for display, replies resulting from the selected interrogation mode. Utilizing the antenna defined in paragraph 6.11.2 the effective reply display azimuth beamwidth shall be a maximum of 15°.

*3.4.1.1 Interrogation - Upon command, the interrogator set shall generate the operator-selected interrogations. The interrogations shall be generated at 1030 MHz. The interrogation shall normally include an Interrogation Side Lobe Suppression (ISLS) control pulse; however, provision shall be made internal to the synchronizer SN-416A/APX-76 for disabling the ISLS control pulse. Interrogations shall be generated for a maximum of five to ten seconds upon each initiation by the operator except as noted in paragraph 3.4.2.3. The interrogation rate shall be timed by a trigger input from the radar and shall be counted down by the interrogator as follows:

Nominal Radar PRF Range	Count Down Ratio
0-400	1:1
400-850	2:1
850-1250	3:1
1250-1650	4:1
1650-2000	5:1

(For prf's above 2200 the SIF and Mode 4 pretriggers will be counted down by at least 2:1 by the trigger delay circuitry in the synchronizer before application to the interrogation countdown circuitry. In no case shall the resulting interrogation rate be greater than 450.)

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3.4.1.1.1 Interrogation Radiation - The interrogator set shall interrogate targets in view of the radar. The normal radiation technique will include a beam sharpening function to control the effective beamwidth of the radiation. The typical antenna shall be designed for radiating sum and difference patterns. The sum pattern shall be a unidirectional pattern and the difference pattern shall produce a minimum (null) at the unidirectional pattern peak.

3.4.1.1.2 Interrogation RF Characteristics - The interrogator set rf characteristics shall conform with the requirements of DOD-AIMS 65-1000 Technical Standard for the ATCRBS/IFF/MARK XII SYSTEMS (AIMS). The normal interrogation peak rf power at the antenna shall not be less than 29 dBW. The normal ISLS pulse peak rf power at the antenna shall not be less than 34 dBW. The ratio between the ISLS and interrogation peak power at the antenna shall be nominally 6 dB. These requirements are not to be construed as a relaxation on requirements of Section 3.5 governing the receiver-transmitter and switch amplifier.

3.4.1.2 Reception - The receiver shall employ receiver side lobe suppression (RSLS) to provide the reply display beamwidth control and shall contain a sum channel and a difference channel which shall normally be connected to the sum and difference patterns of the antenna, respectively. The receiver sum channel shall be duplexed to the transmitter and the difference channel shall be duplexed to the switch-amplifier when the switch amplifier is used.

3.4.1.2.1 Receiver Frequency - The center frequency of the receiver passband shall be nominally 1090 MHz.

*3.4.1.2.2 Receiver Bandwidth - The bandwidth of the receiver, measured between the points 3 dB down from maximum response, shall not be less than 8.0 MHz. The receiver bandwidth, measured between points 40 dB down from maximum, shall not be more than 22 MHz.

3.4.1.2.3 Receiver Sensitivity - In the absence of GTC, the receiver sensitivity shall be capable of being set so that a signal level of no greater than -83 dBm is required to produce decoding of 90% of the Mode 4 replies applied to the sum channel antenna receptacle of the receiver-transmitter. Under all combinations of service conditions a degradation of up to 4 dB is allowable.

3.4.1.2.4 Receiver Gain - There shall be a maintenance adjustable control which will set the gain of the interrogator set receiver. The control shall allow a reduction of the receiver decoding sensitivity of up to 20 dB.

3.4.1.2.5 Gain Time Control - Gain time control shall be utilized. It shall reduce the receiver gain following each interrogation to a low level and then raise the gain exponentially with time matching the expected received signal level. Detailed criteria appear in paragraph 3.5.2.

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3.4.1.3 Display Video - The interrogator set shall decode the replies accepted by the receiver and generate IFF reply video display signals for use with the associated radar indicator for identification of friendly radar targets.

3.4.1.4 System Control - The interrogator set shall be controlled by an operator. Controls are required to initiate operation, select the desired IFF mode, and codes, maintain code changer setting when aircraft power is turned off and to control loop testing (see 3.4.1.6).

3.4.1.5 Performance Monitor Function - The interrogator set shall contain built-in performance monitor circuitry which shall sample critical operating parameters and indicate malfunction. System faults shall be localized to the line replaceable unit by means of fault indicators which shall be provided on the front panel of each major component. Fault alarm indication shall be such as to retain the alarm signal indication in its last state when dc power is removed from the unit. During each interrogation period, the state of operation of the system shall be indicated on the interrogator control unit as determined by combining all the individual unit fault alarm signals.

3.4.1.6 Loop Test - A loop test capability shall be provided by enabling interrogation and display of replies from the onboard transponder to check R-F link and IFF coding operation.

3.4.2 Interrogator Set Inputs

3.4.2.1 Power - The following power sources are required for the interrogator set: (reference 3.3.14)

(1) 115 volts - 400 Hz - single phase

(2) 28 volts - dc

3.4.2.2 Power Control - The interrogator set will become operative when +28 volts dc is supplied to the receiver-transmitter and to the switch-amplifier provided that 115 volts, 400 Hz, single phase is also supplied. In normal operation, power control may be applied by the radar. A test switch may be provided to enable IFF power for ground testing when the radar is off.

*3.4.2.3 Interrogation Control - A grounding function shall be accepted to enable interrogation (challenge enable). Interrogations shall be transmitted and replies received, processed and provided as IFF display video for a maximum of 5 to 10 seconds unless manually terminated in a shorter time by operator release of the interrogate switch. Provision shall be made internal to the synchronizer for disabling the automatic interrogation terminating function.

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*3.4.2.4 Trigger - The interrogator set shall accept triggers with the following characteristics from the radar when terminated in 90 ohms ±10%.

(1)	Polarity	Positive
(2)	Amplitude	7 to 60 volts
(3)	Rise Time	0.05 µs maximum for triggers less than 0.5 µs wide and 0.5 µs maximum for triggers 0.5 µs wide and greater
(4)	Fall Time	1.0 µs maximum
(5)	Duration	0.175 to 5.0 µs
(6)	Repetition rate	100 to 2000 prf when delayed radar trigger output is used and 100 to 5000 prf when delayed radar output is not used.
(7)	DC level	0 ±1.0 volts

3.4.2.5 Suppression - The receiving circuitry of the interrogator set shall be suppressed when the following input is applied to the receiver-transmitter, where it shall be terminated by 1000 ohms:

(1)	Polarity	Positive
(2)	Duration	any
(3)	Amplitude	18 to 50 volts
(4)	Rise Time	10 volts per µs minimum
(5)	Fall Time	peak amplitude to less than 1.0 volt within 10 µs

*3.4.2.6 Mode 4 Code Control - The following Mode 4 code control inputs (to associated equipment) are required:

(1) Landing gear interlock

(2) Refuel hold

(3) Zeroize return

3.4.2.7 Loop Test Control - A grounding function shall be accepted by the interrogator set as a loop test control to allow replies from the collocated transponder to be observed on the radar display when interrogations are enabled.

*3.4.2.8 Long Range Display Enable - The interrogator set shall accept a plus 28 volt signal to increase the spacing of the double pulse video output when the radar indicator is set for a long range display.

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3.4.3 Interrogator Set Outputs

3.4.3.1 IFF Display Video - IFF display video shall be provided having the following characteristics when terminated in 90 ohms ±10%.

(1)	Polarity	Positive
(2)	Amplitude	3 ±1 volt
(3)	Duration	1 ±0.2 µs
(4)	Rise Time	0.2 µs maximum
(5)	Fall Time	0.3 µs maximum
(6)	DC Level	0 ±0.5 volts

*3.4.3.2 Transponder Suppression - The output transponder suppression gate shall have the following characteristics:

(1)	Polarity	Positive
(2)	Amplitude	20 ±5 volts
(3)	Width	100 to 500 µs (adjustable)
(4)	Termination	1,000 ohms in parallel with 500 pF
(5)	Output	Shall be capable of withstanding a dc level of +100 volts maximum.

3.4.3.3 Fault Alarm Indication - A control lead shall provide, during interrogation periods, an indication as determined by the performance monitor function that operation is not satisfactory.

3.4.3.4 Challenge Indication - A control lead shall provide, during interrogation periods, an indication as determined by the performance monitor that operation is satisfactory.

3.4.3.5 Delayed Trigger - The interrogator set shall provide a delayed trigger for operation of a radar. The delayed trigger shall have the following characteristics when terminated in 90 ohms ±10%.

(1)	Polarity	Positive
(2)	Amplitude	50 ±10 volts
(3)	Duration	2.2 ±1 µs
(4)	Rise Time	0.1 µs maximum
(5)	Fall Time	1.5 µs maximum
(6)	DC Level	0 ±1 volt

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3.4.3.6 Radar Trigger Bypass - Plus 28 vdc control power shall be provided to actuate a relay, or equivalent, to insert the trigger delay when the interrogator set is being employed.

3.4.3.7 Audio Challenge Indication - An audio signal shall be provided for the aircraft intercommunication system, during the interrogation period, having the following characteristics:

(1)	Frequency	400 Hz
(2)	Amplitude, adjustable	1 to 8 volts
(3)	Load impedance	10,000 ohms
(4)	Source impedance	600 ohms ±10%

*3.4.3.8 FR Inhibit - A control lead shall provide a ground indication at all times except when the interrogator set is in standby. In standby, this control lead shall be open circuited.

*3.4.3.9 Not Enabled Indication - One control wire and one control return wire shall provide the +28 bus voltage for operation of a remote Not Enabled Indicator when the interrogator set is in standby. When not in standby, the control wire shall be open circuited.

*3.4.3.10 System Interconnection - The system interconnections are detailed in DOD AIMS Drawing No. X66H-1675, Installation Information Wiring Diagram, Interrogator Set AN/APX-76.

3.5 Detailed Requirements

*3.5.1 Control, Interrogator Set - The interrogator control shall be designed in accordance with the requirements of MIL-C-6781, with the exceptions and additions called out herein, and shall fulfill the following requirements.

*3.5.1.1 Control, Interrogator Set C-7383/APX-76(V)

*3.5.1.1.1 Lighting - The lighting shall be in accordance with MIL-P-7788, Class 1-R, Type IV.

3.5.1.1.2 Environment - The interrogator control shall meet the environmental requirements of MIL-E-5400 for Class 2 equipment.

3.5.1.1.3 Interference - The interrogator control shall meet the interference requirements of MIL-I-6181.

3.5.1.1.4 Performance - The control shall provide the following control functions and indications. These control functions and indications shall be in addition to those from external sources listed in section 3.4.

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*3.5.1.1.4.1 Mode Select - One six-position switch shall be provided to select a mode for interrogation and decoding when interrogations are enabled or to place the interrogator set in Standby. The switch shall permit selection of SIF Modes 1, 2, or 3/A, Mode 4 Code A, Mode 4 Code B, or Standby. The switch shall route the disable bus input from the synchronizer back to the synchronizer to enable Modes 1, 2, 3 or 4 (3.5.4.8.1). The switch shall provide a closure between the Mode 4 Code B Selection input and the Mode 4 Code A/B Common input when Mode 4 Code B is selected. The Switch shall cause a closure to ground of the FR Inhibit output, except in Standby. The switch shall cause a closure to the +28 bus voltage of the Not Enabled Indication Output when in Standby.

*3.5.1.1.4.2 Correct Code Challenge Enable - A momentary switch shall be provided to enable interrogator operation by routing the disable bus from the synchronizer back to the synchronizer (3.5.4.2). This challenge function shall allow only correctly coded SIF or Mode 4 replies to be displayed.

*3.5.1.1.4.3 SIF Code Selection - Four in-line, edgewise mounted, thumb wheel switches shall be provided for selection of the SIF code for decoding in the mode selected by the switch of 3.5.1.3.1. Each of these switches shall have eight positions and shall be numbered 0 through 7 consecutively. The illuminated numbers shall appear on a drum-shaped portion of each thumb wheel. Each of the switches shall, depending on the code selected, route the enable and/or disable bus inputs from the synchronizer back to the synchronizer for decoding purposes (3.5.4.4.2).

3.5.1.1.4.4 Loop Test - Facilities shall be provided to allow enabling by a spring return switch of an interrogator-transponder loop test by supplying a ground to the synchronizer (3.5.4.6). This function shall be capable of operation in parallel with an external control circuit.

3.5.1.1.4.5 Fault Indication - An amber indicator light of the press-to-test and turn-to-dim type shall be provided to be operated in addition to a possible external fault indicator.

3.5.1.1.4.6 Challenge Indication - A green indicator light of the press-to-test and turn-to-dim type shall be provided to be operated in addition to a possible external challenge indicator.

*3.5.1.1.4.7 Mode 4 Alarm Override - A switch shall be provided to override the Mode 4 Alarm Function by grounding the Mode 4 Alarm Override input from the Interrogator Computer.

*3.5.1.1.5 Power Requirements - The interrogator control shall require +28 volts dc to operate, and either 5 volts, 400 Hz, single

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phase or 28 volts, 400 Hz, single phase for panel lighting. The total power consumption of the unit shall be 10 watts maximum (see 3.3.14).

*3.5.1.1.6 Dimensions and Weight - The dimensions and weight of the interrogator control shall be in accordance with Table I attached.

*3.5.1.1.7 Controls - The following operational type controls shall be located on the front panel of the interrogator control:

Control	Function
TEST/CHAL CC switch	Shall activate the interrogator transponder loop test on normal challenge when held in TEST position. Shall initiate a challenge with correct code only display for a maximum of 5 to 10 seconds except as noted in paragraph 3.4.2.3 when held in the CHAL CC position.
MODE switch	Shall select a mode for interrogation and decoding when interrogations are enabled, or shall place the equipment in standby.
CODE switches (4)	Shall select the SIF code for decoding in the mode selected by the MODE switch.
M4 ALARM OVERRIDE switch	Allows the Mode 4 alarm to be overridden.

*3.5.1.1.8 Electrical Connections - Connections to external circuits shall be provided as follows:

Reference Designation	Receptacle Type	Function
J1	MS3112E-20-41P	Inlet and outlet for all input and output signals, respectively.

*3.5.1.2 Control, Interrogator Set C-7994/APX-76A(V) - This interrogator control, peculiar to the F-111 aircraft, shall be identical to the C-7383, except that the lighting shall be in accordance with Exhibit SENX-LM-66-1, dated 5 August 1966.

*3.5.1.3 Control, Interrogator Set C-7959/APX-76A(V) - This interrogator control, peculiar to the P3 aircraft, shall be identical to the C-7383, except that the lighting shall be in accordance with MIL-P-7788, Class 2W, Type IV.

*3.5.2 Receiver-Transmitter, Radio RT-868A/APX-76(V) - The receiver-transmitter shall be designed in accordance with MIL-E-5400 for Class 2X equipments, and fulfill the following requirements.

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3.5.2.1 Receiver Function - The receiver shall accept IFF transponder responses from an antenna. The receiver shall operate at a frequency of 1090 MHz. The receiver shall be a dual-channel type which in conjunction with a dual-channel antenna shall provide for receiver-side-lobe suppression (RSLS).

3.5.2.1.1 Sum Channel Performance

*3.5.2.1.1.1 Bandwidth - The response of the receiver as measured at the second detector shall be no more than 3 dB below maximum response over a bandwidth of 8 MHz. The response shall be no more than 3 dB below maximum response at 3.8 MHz and less megahertz above and below 1090 MHz, and shall be no less than 3 dB below maximum response at 5.7 and more MHz above and below 1090 MHz. The bandwidth shall be less than 22 MHz between points 40 dB below maximum response. Points on the average tilt of the selectivity curve measured at ±3 MHz from 1090 shall not differ in amplitude by more than 1 dB. If two or more peaks occur between the points 3 dB down, the dip between any two peaks shall not be greater than 1.0 dB below the level of the lower peak.

3.5.2.1.1.2 Frequency and Bandwidth Stability - The receiver local oscillator shall be at 1030 MHz and shall remain within ±0.2 MHz of this frequency under all combinations of service conditions. The receiver bandwidth specified in 3.5.2.1.1.1 shall hold under all combinations of service conditions.

3.5.2.1.1.3 Sensitivity and Gain

*3.5.2.1.1.3.1 Noise Figure - The receiver shall have a noise figure of 12 dB or better. This characteristic need be demonstrated only during preproduction tests and during sampling or special tests as designated in the contract.

3.5.2.1.1.3.2 Mode 4 Sensitivity - With the internal preset receiver gain control set to limit the squitter rate at the Mode 4 Reply Video Output (3.5.2.2.1) to a maximum of 10 per second, the maximum Mode 4 reply input signal level at the sum receiver input required to produce 90% correct decode shall be -83 dBm. This sensitivity is defined as normal sensitivity. At this input signal level a minimum of 4 volts of signal shall be available at the Receiver Video Output (3.5.2.1.4) for use by the synchronizer or other SIF decoding equipment of PPI displays. Under all combinations of service conditions a degradation of up to 4 dB is allowed.

3.5.2.1.1.3.3 Tangential Sensitivity - The receiver shall have a tangential sensitivity, measured at the video output of the IF amplifier (IF Video), of -86 dBm or better.

*3.5.2.1.1.4 Spurious Response - The image rejection of the sum channel of the receiver shall be at least 70 dB. At all

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frequencies above and below a receiver bandwidth of 36 MHz the response to any signal shall be at least 70 dB below the response to a signal at the center frequency of the sum channel of the receiver.

3.5.2.1.1.5 Receiver Gate - The receiver shall provide a receiver gate which, when actuated by an external GTC trigger, shall return the receiver sensitivity to normal from a desensitized level of -12 dBm or less within a period of 25 microseconds when the GTC circuitry is not enabled. The length of this gate shall be adjustable throughout the minimum range of 300 to 6200 microseconds measured from the trigger pulse. Jitter on the trailing edge of the gate shall not exceed 1% of the gate duration. At the end of the gate, the receiver sensitivity shall return to the desensitized level within a period of 75 microseconds.

3.5.2.1.1.6 Gain Time Control - The receiver shall provide gain time control (GTC) circuitry which, when the receiver gate is triggered, shall cause the receiver sensitivity to rise from the desensitized level to an adjustable initial or GTC level that is from 20 dB to 40 dB less sensitive than normal sensitivity within a period of 15 microseconds measured from the receipt of the GTC trigger by the receiver gate. The receiver sensitivity shall then increase from this initial level at 15 microseconds after the receipt of the GTC trigger to within 3.0 dB of normal sensitivity in such a manner as to be within 2 dB of a 6-dB-per-octave rate-of-change curve under normal room ambient conditions. Under extreme environmental test conditions, the receiver sensitivity shall increase in such a manner as to be within 3 dB of a 6-dB-per-octave rate-of-change curve. The first octave shall be from 15 to 30 microseconds after the receipt of the GTC trigger. The GTC circuitry shall be selectable by means of a switch on the front panel.

3.5.2.1.1.7 Recovery Time - When the GTC gate circuit is disabled, the receiver shall recover from the effects of a transmitted signal to within 3 dB of normal sensitivity in not more than 15 microseconds measured from the trailing edge of the transmitter pulse.

*3.5.2.1.1.8 Anti-jamming - When set for normal operation, the receiver shall be capable of operating in the presence of jamming as follows:

(1) CW: Pulsed signals greater than -71 dBm shall produce useful video outputs in the presence of a CW jamming signal of -56 dBm or less.

(2) MCW: In the presence of CW modulated 30 percent at 1000 Hz at a level of -49 dBm, the video output shall not contain more than one volt peak-to-peak envelope of 1000 Hz and its harmonics.

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3.5.2.1.1.9 Circuitry - The use of circuits which impair equipment operation in the presence of railing and FM jamming shall be avoided whenever practicable. The anti-jamming circuits used shall not impair normal operation of the receiver in the absence of jamming.

3.5.2.1.2 Difference-Channel Performance

*3.5.2.1.2.1 Receiver - The receiver used in the difference channel shall be similar to that used in the sum channel. It shall meet the requirements of paragraphs 3.5.2.1.1.1, 3.5.2.1.1.2, 3.5.2.1.1.4, 3.5.2.1.1.8 and 3.5.2.1.1.9.

3.5.2.1.3 General Performance

3.5.2.1.3.1 Amplifiers - The center frequency of the I-F amplifiers in the sum and in the difference channels of the receiver shall be nominally 60 MHz.

3.5.2.1.3.2 Receiver Protection - The receiver and all other components of the receiver-transmitter shall not be damaged by the transmitter power, or external signals up to 10 volts peak and an average power of +20 dBm in amplitude applied to the antenna terminal.

3.5.2.1.3.3 Local Oscillator Isolation - The signal delivered by the local oscillator into the antenna terminated by a 50 ohm resistive load shall not be stronger than -67 dBm under all combinations of service conditions.

3.5.2.1.3.4 Receiver Overload - Receiver overload is the temporary or extended desensitization of the receiver following receipt of a strong signal, or block of signals. No appreciable receiver overload shall occur for input signals of 60 dB above Normal Sensitivity or -20 dBV, whichever is the smaller. Receiver operation in the presence of overload, or imminent overload is more fully defined by 3.5.2.1.1.7 (Recovery) and 3.5.2.1.4.2 (Fidelity).

3.5.2.1.3.5 Frequency Adjustment and Test Points - Tuning adjustment and suitable test points required for alignment of the receiver shall be readily accessible for internal service adjustment.

3.5.2.1.4 Receiver Video Response and Output - The design of the receiver and the receiver video output circuitry shall preserve to the greatest extent possible the information content of the received pulse trains and present the receiver output to panel jacks. Receiver noise shall normally be reduced by means of a base clipping threshold set at normal sensitivity.

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3.5.2.1.4.1 Receiver Video Output - The receiver video output shall present the undecoded output. The level of the standard video output shall be limited to be 5 ±1 volt when the output is terminated in 75 ±10 percent ohms.

3.5.2.1.4.2 Video Fidelity

(1) Reply signals within the range of 3 to 50 dB above Normal Sensitivity shall exhibit minimum degradation of rise time and fall time. Rise and fall time shall not exceed 0.08 and 0.12 microseconds, respectively. The shift in spacing of the pulses in any reply train shall be less than 0.05 microsecond and the width shall not vary more than 0.1 microsecond from received pulse width as controlled by DOD-AIMS 65-1000.

(2) Interleaved pulse trains spaced 0.7 microseconds at the leading edges and having any level differential from 0 to at least [50 dB minus (Instantaneous GTC attenuation +3) dB] or 30 dB whichever is smaller, shall be processed with less than ±0.05 microsecond pulse position shift within either train. The following conditions shall also apply:

(a) Reply trains may be identical to any combination of standard replies.

(b) Pulse width and spacings of the reply trains may be nominal.

(c) The weaker signal shall be at least 3 dB above normal sensitivity as specified in paragraph 3.5.2.1.1.3.

(d) The stronger signal shall not exceed -33 dBm.

(3) These characteristics apply under all combinations of standard reply signals and noise up to, but not exceeding, duty cycles of 25% as averaged over one receiver gate period.

(4) The peak-to-peak amplitude of hum, crosstalk and switching transients in the video output shall not exceed 30 dB below peak signal level.

*3.5.2.1.5 RSLS Performance - The receiver-transmitter shall have an RSLS function. Two receiver input channels, the normal sum channel and the difference channel, shall be provided with the

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relative level of the inputs determining the presence of the normal receiver output. The normal receiver output shall be gated off unless the sum input exceeds the difference input by a ratio preset to a value settable over the range of 6 to 15 dB. This performance shall be provided over a signal input variation extending from a sum channel signal level that exceeds the specified normal sensitivity of paragraph 3.5.2.1.1.3.2 by the preset ratio plus 4 dB to a signal level of -33 dBm. The RSLS performance shall be provided at a signal frequency of 1090 MHz and under Standard Conditions, paragraph 3.3.11, with no greater than a ±4 dB change in the sum and difference channel input signal ratio permitted.

*3.5.2.2 Mode 4 Video Function - The receiver-transmitter shall delay-line decode the triple pulse Mode 4 replies developed at the receiver output and provide the decoded Mode 4 Reply Video as an output. This unit shall accept a Mode 4 ISLS trigger, delay it and shape it to modulate the transmitter and delay it and shape it to provide a Mode 4 ISLS Gate.

*3.5.2.2.1 Mode 4 Reply Video Output - Mode 4 reply signals as controlled by DOD-AIMS 65-1000 shall be decoded using a tapped delay-line. A single unmodulated pulse of any width shall not be decoded as a Mode 4 reply. Mark X (SIF) replies shall not be decoded as Mode 4 replies. The decoded Mode 4 Reply Video shall be provided as an output with the following characteristics when terminated in 90 ±10% ohms:

(1)	Amplitude	+3.25 ±1.75 volts
(2)	Duration	0.5 ±0.1 μs
(3)	Rise Time	0.1 μs maximum
(4)	Fall Time	0.15 μs maximum
(5)	Quiescent signal	±0.5 V (dc + undesired signal maximum)

3.5.2.2.2 Mode 4 ISLS Trigger Input - The receiver-transmitter shall accept as an input the Mode 4 Side Lobe Suppression trigger output of the interrogator computer as controlled by the Mode 4 Input/Output Data, DOD-AIMS 64-900.

3.5.2.2.3 Mode 4 ISLS Transmitter Output - The transmitter shall transmit the Mode 4 ISLS pulse within the requirements of paragraph (3.5.2.3.3) when the Mode 4 ISLS trigger pulse is properly received from the interrogator computer.

*3.5.2.2.4 Mode 4 ISLS Gate Output - The receiver-transmitter shall provide as an output a Mode 4 ISLS Gate for use by the rf switch driver. The gate shall have the following characteristics when terminated by 90 ±10% ohms:

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(1)	Amplitude	+5 ±1 volts
(2)	Duration	1.5 ±0.1 μs
(3)	Spacing	1.4 ±0.1 μs after Mode 4 ISLS Trigger Input
(4)	Rise Time	0.1 μs maximum
(5)	Fall Time	0.15 μs maximum
(6)	Quiescent signal	±0.5 V (dc + undesired signal maximum).

3.5.2.3 Transmitter Function - The transmitter shall accept coded video pulse trains from an external source and the internally generated Mode 4 ISLS pulse and convert the coded video pulse trains and Mode 4 ISLS pulse to radio-frequency pulse groups for transmission as IFF interrogations. The transmitter shall be capable of generating all standard SIF and Mode 4 interrogations.

3.5.2.3.1 Transmitter Frequency - The transmitter shall be designed for operation at 1030 ±0.2 MHz under all combinations of service conditions and operation including mismatching of up to 1.6:1 voltage SWR at the antenna connections and over any variation of interrogation rate up to a maximum duty cycle of 1 percent.

*3.5.2.3.2 Transmitter Power Output - The transmitter shall be capable of delivering not less than 2000 watts, peak rf power (as averaged over an rf pulse) to the antenna front panel jack over the entire range of environmental conditions specified herein at a frequency of 1030 ±0.2 MHz and at any duty cycle up to a maximum of 0.6 percent. Under standard conditions (para. 3.3.11) the specified operation at a duty cycle up to a maximum of 1 percent shall be provided. Operating tests for demonstration of equipment life and reliability shall be carried out at a maximum duty cycle of 0.6 percent. A front panel adjustment shall be provided capable of reducing the transmitter power output in 2 ±1 dB increments in two steps in order to be able to suit the needs of a specific installation. This reduction in transmitter power shall be accomplished without degradation of receiver sensitivity. The transmitter performance shall not be affected by hot switching of an associated rf switch.

3.5.2.3.3 Modulation - The transmitter shall be capable of generating standard rf interrogations, as controlled by DOD-AIMS 65-1000, in response with paragraph 3.5.4.3.1. The amplitude droop of any rf interrogation train shall be held to less than 1 dB.

3.5.2.3.4 Jitter - The jitter of the rf pulse shall not be more than ±0.03 microsecond with respect to the corresponding point on the leading edge of the initial and effective trigger pulse.

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3.5.2.3.5 Transmitter Delay - Transmitter delay measured from the minimum voltage point on the modulation input pulse which causes triggering of the transmitter to the 50 percent amplitude point of the output rf pulses shall not exceed 1.0 microsecond. Variation between equipments shall not exceed ±0.05 microsecond.

*3.5.2.3.6 RF Probe - A directional rf probe shall be provided in the sum channel line between the duplexer and the antenna terminal. The probe shall terminate in a front panel BNC Jack and shall produce a standing wave ratio of not more than 2 dB in a 50 ohm line connected to it. The probe shall not cause a standing wave ratio of more than 0.5 dB in the final duplexer-to-antenna line. When the "SUM ANTENNA" jack is terminated with a 50 ohm resistive load, the power appearing at the probe shall be 20 ±1.0 dB less than that flowing toward the load, and shall be at least 40 dB less than that flowing away from the load. The forward attenuation of the probe in dB at 1030 MHz shall be marked on the front panel adjacent to the BNC jack. This attenuation marking shall be accurate to ±0.5 dB.

3.5.2.3.7 Transponder (XP) Suppression Gate Output - The receiver-transmitter shall provide for the generation and delivery of an XP Suppression Gate for the purpose of desensitizing Radar Identification Set AN/APX-64 or equivalent. The XP Suppression Gate shall be generated for each interrogation transmitted and shall precede the first rf interrogation pulse by at least 0.1 microsecond. Adjustment of the XP Suppression Gate duration shall be supplied in the receiver-transmitter unit. The XP Suppression Gate shall have the following characteristics when applied to a load consisting of nominal 1000 ohms resistance in parallel with 500 micro-microfarads capacitance:

(1) Amplitude — 20 ±5 volts

(2) Polarity — Positive

(3) Duration (measured at 50% amplitude) 100 to 500 microseconds. This adjusted value shall hold with ±5% over all combinations of service and environmental conditions.

(4) Rise Time — Less than 0.1 microsecond

(5) Fall Time — Shall not exceed 2.0 microseconds

Jitter on trailing edge shall not exceed 1.0 microsecond. When this terminal is shorted, the equipment shall not be damaged.

3.5.2.4 Interrogator (IR) Suppression Gate Input - The receiver-transmitter shall accept the IR Suppression Gate specified in (3.4.2.5) or equivalent. The Receiver Video Output (3.5.2.1.4.1) and Mode 4 Reply Video Output (3.5.2.2.1) shall be gated off when the IR Supression Gate is present. The IR Supression Gate input shall be terminated with 1000 ohms.

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*3.5.2.5 Loop Test Provisions - When loop test is enabled by a ground from the interrogator control, the receiver-transmitter shall function normally except that the Mode 4 ISLS Functions (3.5.2.2.3) and (3.5.2.2.4), the RSLS function (3.5.2.1.5), the XP Suppression Gate Output (3.5.2.3.7) and the IR Suppression Gate Input (3.5.2.4) shall be disabled.

3.5.2.6 Performance Monitor Function - A built-in performance monitor shall sample critical operating parameters and indicate malfunction by means of fault indicators which shall be provided on the front panel of the receiver-transmitter. Alarm indication shall be such as to retain the alarm signal indication in its last state when dc power is removed from the unit. Three separate indicators shall be provided to indicate Transmitter, Receiver, and Video faults to enable fault localization. When a fault exists in the receiver-transmitter, a grounding function shall be supplied as an output to the synchronizer where a signal shall be developed to indicate a fault in the system.

3.5.2.6.1 Actuation - The built-in performance test shall be carried out once each IFF repetition period whenever the challenge function is initiated and the receiver gate is enabled.

3.5.2.6.2 Transmitter Operation Checks - The receiver-transmitter shall monitor the following transmitter operations during the challenge period 3.5.4.2 if a GTC trigger is present:

(1) Normal transmitter power output present for each modulation input pulse. The GO/NO-GO power threshold shall be adjustable from 250 W to 1000 W when the power switch is in high power. The transmitter fault indicator shall indicate full "GO" when power is 3 dB above the setting and full "NO-GO" when power is 3 dB below the setting.

(2) Low voltage power supply voltage levels.

*3.5.2.6.3 Receiver Operational Checks - The receiver-transmitter shall monitor the receiver operation during the challenge period 3.5.4.2, if a GTC trigger is present, as described below:

(1) Signal Generator Level:

The self-test signal generator shall inject a signal from -62 to -68 dBm into the input cavities of the sum and difference channel preselectors as measured at the sum and difference channel antenna ports respectively. The rf power output level of the signal generator shall remain within ±3 dB from its initial value.

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(2) A threshold circuit shall be employed at the IF sum video line and its GO/NO-GO point shall be adjustable over a 15 dB range. This threshold circuit shall operate on the self-test signal generator pulse so that the equivalent RF level of the signal generator can be reduced by from 0 to 15 dB. (i.e. If the signal generator has an actual RF output level of -65 dBm as defined in (1) then the adjustment of the threshold circuit shall be capable of converting this signal to equivalent rf levels of from -65 to -80 dBm based on the minimum signal amplitude of IF sum video pulses necessary for proper mode 4 decoding.)

(3) The Receiver fault indicator shall always show a full GO (Black) indication when the input signal to the IF sum video threshold circuit is increased by 2 dB from the GO/NO-GO setting. The Receiver fault indicator shall always show a NO-GO (Red) indication when the input signal is reduced by 2 dB from the GO/NO-GO threshold setting.

3.5.2.6.4 Video Operational Checks - The receiver-transmitter shall monitor the following video operations during the challenge period of 3.5.4.2 if a GTC trigger is present:

(1) Normal receiver video amplifier operation. The GO/NO-GO point shall be adjustable from 0.5 to 2 volts. The video fault indicator shall show full GO when a pulse appearing during self-test time is 5% greater than the setting and full NO-GO when such a pulse is 5% less than the setting.

(2) Normal transmitter modulation present for input trigger pulses.

3.5.2.6.5 Self-Test Gate Output - The self-test gate shall occur at the end of the receiver gate, shall be available as an output at pin C of the front panel test connector and have the following characteristics when terminated in 1000 ohms in parallel with 1000 pF:

(1)	Polarity	Positive
(2)	Amplitude	3.5 to 6.0 volts
(3)	Rise Time	2.5 µs maximum
(4)	Fall Time	0.5 µs maximum
(5)	Quiescent signal	0 to 0.6 volts

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*3.5.2.7 Power Requirements - The receiver-transmitter shall require +28 volts dc and 115 volts, 400 Hz, single phase to operate. The total power consumption of the unit shall be 190 watts maximum. (See 3.3.14.)

*3.5.2.8 Dimensions and Weight - The dimensions and weight of the receiver-transmitter shall be in accordance with Table I. attached.

*3.5.2.9 Controls - The following controls shall be provided on the receiver-transmitter:

Control	Location	Function
POWER ADJ switch	Front Panel	Shall reduce the power output in two steps of 2 ±1 dB each.
GTC ON/OFF switch	Front Panel	Shall enable or disable the gain time control circuits of the receiver.
RCVR GATE ON/OFF switch	Front Panel	Shall enable or disable the receiver gate circuits.
AC circuit breaker	Front Panel	Shall control primary ac power to the receiver-transmitter.
DC circuit breaker	Front Panel	Shall control primary dc power to the receiver-transmitter.
PUSH TO TEST switch	Front Panel	Shall create artificial faults to actuate three fault indicators on the front panel.

*3.5.2.10 Electrical Connections - Connections to external circuits shall be provided as follows:

Reference Designation	Receptacle Type or Approved Equivalent	Function
J1	DS00-61-21P (Deutsch)	IR suppression gate, Mode 4 ISLS trigger, GTC trigger, and interrogator trigger inlets; receiver video, Mode 4 reply video, Mode 4 ISLS gate, and XP suppression gate outlets.
J2	MS3113H12-10PW	Loop test enable and power inlet. Equipment fault outlet.

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Reference Designation	Receptacle Type or Approved Equivalent	Function
J3	4123-0002 (Solitron)	Sum channel inlet and outlet.
J4	4123-0002 (Solitron)	Difference channel inlet
J5	MS3112E16-26S	Test Connector
J6	819-B3800B-75 (Plessey)	Test Connector

*3.5.3 Switch-Amplifier SA-1568A/APX-76(V) - The switch-amplifier shall be designed in accordance with MIL-E-5400 for Class 2 equipment, and fulfill the following requirements.

3.5.3.1 Switch Function - The switch-amplifier shall normally pass the interrogation RT pulses (3.5.2.3) generated in the receiver-transmitter to the sum mode input of the interrogator antenna. Upon receipt of an ISLS control gate (3.5.4.3.3) from the synchronizer, the switch-amplifier shall pass the corresponding ISLS input rf signal via its amplifier to the difference mode input of the interrogator antenna.

3.5.3.1.1 Pulse Characteristics - The rise time, fall time, and width of the output rf pulses shall be in accordance with DOD-AIMS 65-1000.

*3.5.3.1.2 Interrogation Insertion Loss - Except during the ISLS period, the switch-amplifier shall have a maximum insertion loss of 1.8 dB.

*3.5.3.1.3 Input VSWR - The Voltage Standing Wave Ratio measured looking into the RT sum input shall not be greater than 1.6 to 1 when the outputs are terminated in 50 ohms.

3.5.3.2 Amplifier Function - During the ISLS period the switch-amplifier shall amplify the interrogation rf signal and apply it to the difference mode input of the interrogator antenna. The duty cycle of the switch-amplifier shall not exceed 0.03 percent.

3.5.3.2.1 Gain - With an rf signal input level of 30 to 34 dBW, the output shall be between 4 and 7 dB greater than the input.

3.5.3.2.2 Pulse Characteristics - The limits of rise time, fall time, and width of the output pulse shall be in accordance with the requirements of DOD-AIMS 65-1000.

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*3.5.3.3 Interrogation Isolation - At the sum and difference antenna ports of the switch-amplifier, the isolation, defined as the difference in power level between the peak pulse power of the desired pulse(s) and the pulse power level of the undesired pulse(s) shall be 20 dB minimum.

3.5.3.4 Reply Function - The switch-amplifier shall couple the sum and difference antenna replies to the respective sum and difference channels of the receiver-transmitter.

3.5.3.4.1 Reply Isolation - The difference in power level between an output present at the sum or difference antenna port due to a difference or sum antenna reply, respectively, shall be 30 dB minimum.

*3.5.3.4.2 Reply Insertion Loss - The maximum sum channel insertion loss at 1090 ±5 MHz shall be 1.8 dB. The maximum difference channel insertion loss shall be 2.0 dB at 1090 MHz and 3.0 dB at 1085 and 1095 MHz.

3.5.3.5 Power Control Function - The switch-amplifier shall switch 115 volts, 400 Hz, single phase to provide power to the interrogator computer and to the synchronizer when the Power Control input (3.4.2.2) is present. If the switch amplifier is not used in the system power control must be provided externally.

3.5.3.6 Performance Monitor Function - A built-in performance monitor shall sample all critical operating parameters and indicate any malfunctions by means of a fault indicator which shall be provided on the front panel of the switch-amplifier. Alarm indication will be such as to retain the alarm signal indication in its last state when power is removed from the unit. The unit shall also provide the following signals to the synchronizer:

Condition	Signal
Fault	0 ±1 vdc into a minimum of 200 ohms
No Fault	Between +5 vdc and +90 vdc into 50 K ohms

3.5.3.6.1 Required Operational Checks - The switch-amplifier shall monitor the following operations during the challenge period:

(1) For each normal transmitter pulse received from the receiver-transmitter, not accompanied by a corresponding ISLS trigger pulse, a check shall be made to determine the presence of a pulse incident at the sum antenna jack.

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(2) For each normal transmitter pulse received from the receiver-transmitter, accompanied by a corresponding ISLS trigger pulse, a check shall be made to determine the presence of an amplified pulse incident at the difference antenna jack.

(3) The presence of the dc voltages in the low voltage power supply shall be determined.

*3.5.3.7 Power Requirements - The switch-amplifier shall require 28 volts dc and 115 volts, 400 Hz, single phase to operate. The total power consumption of the unit shall be 30 watts maximum. (See 3.3.14.)

*3.5.3.8 Dimensions and Weight - The dimensions and weight of the switch-amplifier shall be in accordance with Table I attached.

3.5.3.9 Controls - The following controls shall be provided on the switch-amplifier:

Control	Location	Function
AC circuit breaker	Front Panel	Shall control primary ac power to the switch-amplifier.
DC circuit breaker	Front Panel	Shall control primary dc power to the switch-amplifier.

*3.5.3.10 Electrical Connections - Connections to external circuits shall be provided as follows:

Reference Designation	Receptacle Type or Approved Equivalent	Function
J1	MS3114H12C10P	Power inlet and outlet; fault indication outlet.
J2	4087-0003 (Solitron)	ISLS control gate inlet.
J3	4087-0003 (Solitron)	Sum channel inlet and outlet to and from receiver-transmitter.
J4	4072-0010 (Solitron)	Difference channel outlet to receiver-transmitter.
J5	4072-0010 (Solitron)	Sum channel inlet and outlet to and from antenna.
J6	4072-0010 (Solitron)	Difference channel inlet and outlet to and from antenna.

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*3.5.4 Synchronizer, Electrical, SN-416A/APX-76(V) - The synchronizer shall be designed in accordance with MIL-E-5400 for Class 2 equipment and shall fulfill the following requirements:

*3.5.4.1 Radar Trigger Function - In normal operation, the synchronizer shall accept triggers from a radar, at a prf of 100 to 2000 and generate output triggers delayed by 462 ±0.6 microseconds to trigger the radar modulator, or shall accept radar pretriggers 462 ±0.6 microseconds before radar T_0 or other specified fixed time reference at a prf of 100 to 5000. Under loop-test conditions (3.5.4.6) the delayed radar trigger shall be generated 462 ±3.0 microseconds (462 ±6.0 µs over temperature extremes) after the reception of the input trigger. During a challenge period (3.5.4.2) a 28 volt dc level shall be supplied by the synchronizer to actuate a radar trigger bypass relay in the radar modulator or any other desired control function.

*3.5.4.1.1 Input Radar Trigger - The synchronizer shall accept triggers or pretriggers from a radar with the following characteristic when terminated in 90 ±10% ohms.

(1)	Polarity	Positive
(2)	Amplitude	7 to 60 volts
(3)	Rise Time	0.05 µs maximum for triggers less than 0.5 µs wide and 0.5 µs maximum for triggers 0.5 µs wide and greater
(4)	Fall Time	1.0 µs maximum
(5)	Duration	0.175 to 5.0 µs
(6)	Repetition rate	100 to 2000 prf when delayed radar trigger output is used and 100 to 5000 prf when delayed radar output is not used.
(7)	DC Level	0 ±1.0 volts

3.5.4.1.2 Delayed Radar Trigger - The delayed radar trigger shall have the following characteristics when terminated in 90 ±10% ohms:

(1)	Polarity	Positive
(2)	Rise Time	0.1 µs maximum
(3)	Fall Time	1.5 µs maximum
(4)	Duration	2.2 ±1 µs
(5)	Amplitude	50 ±10 volts
(6)	DC Level	0 volts ±1.0 volt

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*3.5.4.1.3 Radar Trigger Bypass - The radar trigger bypass control output shall be derived from the 28 vdc bus and have the following characteristics:

(1)	Polarity	Positive
(2)	Amplitude	28 vdc bus voltage
(3)	Output impedance	10 ohms maximum
(4)	Current	200 milliamperes

*3.5.4.2 IFF Pretriggers - When a challenge is enabled, a ground shall be applied to the challenge enable or loop test enable input, or a -4V signal shall be applied to the correct code challenge enable input. The incoming triggers from the radar shall normally produce IFF pretriggers at the prf specified in (3.4.1.1) for a maximum challenge period of from 5 to 10 seconds. However, provision shall be made on the appropriate printed circuit card assembly for disabling the automatic interrogation terminating function. Continuous challenge operation of the AN/APX-76 system shall be possible by applying a 28-volt dc level to the test challenge enable input of the synchronizer from an external source. Grounding of the loop test enable input shall cause SIF and Mode 4 Pretriggers to be delayed by 50 ±2.4 microseconds (50 ±5.4 µs over temperature extremes) from their normal positions with respect to the incoming radar trigger.

3.5.4.2.1 SIF Pretriggers - When an SIF Mode 1, 2 or 3/A is enabled for challenge (3.5.4.8) SIF pretriggers shall be generated to trigger the internal coder circuits. Internal adjustment provisions shall be included to adjust the delay of the SIF pretrigger such that decoded SIF replies will occur at the same range as the nominal position of decoded Mode 4 replies.

*3.5.4.2.2 Mode 4 Pretriggers - When Mode 4 is enabled for challenge, Mode 4 pretriggers shall be generated nominally 9.5 microseconds after the radar trigger to trigger the interrogator computer. Internal adjustment provisions shall be included to adjust the delay of this output trigger relative to the incoming radar such that the first pulse of a display video pulse pair (3.5.4.5) for Mode 4 can be positioned on the display from approximately 15 microseconds earlier to approximately 10 microseconds later (long range display) than its associated radar return for a 462 microsecond pretrigger (3.5.4.1). The Mode 4 pretriggers shall have the following characteristics when terminated in 90 ±10% ohms:

(1)	Polarity	Positive
(2)	Amplitude	1.5 to 5 volts

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(3)	Duration	0.5 to 10.0 μs
(4)	Rise Time	0.10 μs maximum
(5)	Fall Time	5.0 μs maximum
(6)	Undesired signal plus quiescent dc voltage	±0.5 volts peak
(7)	Delay Tolerance	±0.6 μs

3.5.4.3 Coder Functions - When initiated by the IFF pretriggers, the coder circuits shall generate the following outputs:

3.5.4.3.1 Interrogation Triggers - Interrogation trigger outputs shall be generated for modulating the transmitter of the receiver-transmitter.

*3.5.4.3.1.1 SIF Interrogation Triggers - When Mode 1, 2 or 3/A is enabled for challenge, the internal code circuits shall generate SIF interrogation triggers having the correct pulse spacing and duration for modulating the receiver-transmitter. The P_2 pulse shall also normally be generated 2 ±0.1 microseconds following the P_1 pulse, and shall be mixed with the interrogation output pulses. However, provision shall be made on the appropriate printed circuit card assembly for disabling the P_2 pulse from the interrogation. The SIF interrogation trigger pulses shall have the following characteristics when terminated in 90 ±10% ohms:

(1)	Polarity	Positive
(2)	Amplitude	5.0 ±1.0 volts
(3)	Duration	0.8 ±0.05 μs
(4)	Rise Time	0.05 μs maximum
(5)	Fall Time	0.1 μs maximum
(6)	Spacing	
	(a) Mode 1	3 ±0.07 μs
	(b) Mode 2	5 ±0.17 μs
	(c) Mode 3/A	8 ±0.17 μs

3.5.4.3.1.2 Mode 4 Interrogation Triggers - Upon receiving Mode 4 challenge video defined in DOD-AIMS 64-900A from the interrogator-computer, the coder shall produce Mode 4 interrogation triggers having the same pulse spacing and the following characteristics when terminated in 90 ±10% ohms:

(1)	Polarity	Positive
(2)	Amplitude	5.0 ±1.0 volts

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(3)	Duration	0.5 ±0.05 μs
(4)	Rise Time	0.05 μs maximum
(5)	Fall Time	0.1 μs maximum

3.5.4.3.2 GTC Trigger - A gain time control trigger output shall be generated for triggering the receiver GTC circuit of the receiver-transmitter.

3.5.4.3.2.1 SIF-GTC Trigger - For Modes 1, 2 and 3/A, the coder circuits shall generate the GTC trigger within ±0.50 microseconds from the time of the P_3 pulse. The GTC trigger shall have the following characteristics when terminated in 90 ±10% ohms:

(1)	Polarity	Positive
(2)	Amplitude	5.0 ±1.0 volts
(3)	Duration	0.3 to 1.8 μs
(4)	Rise Time	0.1 μs maximum
(5)	Fall Time	0.2 μs maximum

3.5.4.3.2.2 Mode 4-GTC Trigger - Upon receiving a Mode 4-GTC trigger defined in DOD-AIMS 64-900A from the interrogator-computer, the coder shall generate a GTC pulse with the following characteristics when terminated in 90 ±10% ohms:

(1)	Polarity	Positive
(2)	Amplitude	5.0 ±1.0 volts
(3)	Duration	0.3 to 1.8 μs
(4)	Rise Time	0.1 μs maximum
(5)	Fall Time	0.2 μs maximum

3.5.4.3.3 ISLS Control Gate - A control gate shall be generated to enable the rf switch driver circuits in the switch-amplifier to provide the ISLS function.

3.5.4.3.3.1 SIF-ISLS Control Gate - For Modes 1, 2, and 3/A the ISLS control gate shall occur 1.4 ±0.1 microseconds from the start of the interrogation P_1 pulse. The ISLS control gate shall have the following characteristics when terminated in 90 ±10% ohms:

(1)	Polarity	Positive
(2)	Amplitude	5.0 ±1.0 volts
(3)	Duration	1.5 ±0.1 μs
(4)	Rise Time	0.1 μs maximum
(5)	Fall Time	0.15 μs maximum

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3.5.4.3.3.2 Mode 4 ISLS Control Gate - Upon receiving a Mode 4 ISLS gate defined in 3.5.2.2.4 from the receiver-transmitter, the coder shall generate an ISLS control gate with the following characteristics when terminated in 90 ±10% ohms:

(1)	Polarity	Positive
(2)	Amplitude	5 ±1 volt
(3)	Duration	Mode 4 ISLS gate duration (Ref. 3.5.2.2.4) +0.1, -0.05 µs
(4)	Rise Time	0.1 µs maximum
(5)	Fall Time	0.15 µs maximum
(6)	Delay	65 ±15 ns

3.5.4.3.4 Challenge Indication - A grounding signal (+3 volts dc or less) for operation of a 28 volt indicator light (nominal current drain of 40 mA) on the interrogator control shall be provided during the interrogation period if operation is satisfactory as determined by the self-test mixer (3.5.4.7.2).

3.5.4.3.5 Audio Challenge Indication - Upon reception of a ground signal from a source external to the system, the synchronizer shall provide an audio signal for the aircraft intercommunication system during the interrogation period having the following characteristics:

(1)	Frequency	400 Hz
(2)	Amplitude, Adjustable	1 to 8 volts rms
(3)	Load impedance	10,000 ohms
(4)	Source impedance	600 ohms (approximately)

3.5.4.4 Decoder Functions - When enabled by a challenge in Mode 1, 2, or 3/A the decoder shall accept video signals as defined in (3.5.2.1.4) from the receiver of the receiver-transmitter. The decoder shall reject pulses which are less than 0.2 microseconds in duration. After processing the incoming pulses, the decoder shall apply them to a tapped decoding delay line. The receiver video shall be locked out from 412 microseconds after the input radar trigger until the GTC trigger. The decoder shall supply IFF video to the display video mixer.

3.5.4.4.1 Bracket Decoder - The bracket decoder shall recognize a pair of pulses spaced 20.3 ±0.1 microseconds apart as SIF bracket or framing pulses and generate an output for use by the display video mixer circuit (3.5.4.5).

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3.5.4.4.2 Correct Code Decoder - The decoder shall examine the taps of the decoding delay line at the time of the bracket decode to determine whether the reply code of the recognized SIF pulse train agrees with the setting of the SIF code switches on the interrogator control. If the code is correct (i.e. all the required pulses are present and no pulse positions are filled which should be vacant), the decoder shall generate a pulse output for use by the display video mixer circuits.

3.5.4.5 Display Video Mixer - The display video mixer circuits shall generate IFF video for use by the radar display. This video shall be inhibited for non-IFF repetition periods.

*3.5.4.5.1 SIF-IFF Video - For Modes 1, 2 and 3/A the display video mixer shall combine the outputs from the bracket decoder and correct code decoder, if any, and generate a single or double pulse output between 7.5 and 9.5 µs after the last framing pulse of a reply is applied to the decoder. If a long range display enable input is present, the single or double pulse output shall be generated within 1.5 µs after the last framing pulse of a reply is applied to the decoder. When an output pulse is received from the bracket decoder and no pulse is received from the correct code decoder, a single pulse output shall be generated for use by the radar display except during a correct code only challenge as described in paragraph 3.5.4.5.1.1. When pulse outputs are received from the bracket and correct code decoders, the display video mixer shall generate a double pulse output for the radar display. The spacing of the pulses shall be 11.5 to 12.5 µs, if no long range display enable input is received and 26.5 to 29.0 µs, if a long range display enable input is received. The video pulse shall have the following characteristics when terminated in 90 ±10% ohms.

(1)	Polarity	Positive
(2)	Amplitude	2 to 4 volts
(3)	Duration	0.8 to 1.2 µs
(4)	Rise Time	0.2 µs maximum
(5)	Fall Time	0.3 µs maximum

3.5.4.5.1.1 Correct Code Only Display - Upon initiating an external circuit closure, applying -4v control voltage, the SIF-IFF video display signal output from Display Video Mixer shall only consist of the double pulse output indicating a correct code reply (i.e. no pulse output will be present due to bracket decode).

*3.5.4.5.1.2 Long Range Display Enable - Upon receipt of a plus 28 vdc signal (usually obtained from the radar indicator or radar set control) signifying that the radar indicator is set for a Long Range Display, the IFF Video double pulse display spacing shall increase from 12 microseconds to 28 microseconds.

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*3.5.4.5.2 Mode 4 - IFF Video - When Mode 4 is enabled for challenge, the display video mixer shall accept Mode 4 time-decoded video as defined in DOD-AIMS 64-900A and between 7 and 9 µs, provide the first pulse of a double pulse output to the radar display. If a long range display enable is present, the video mixer shall provide the first pulse of a double pulse display within 1 µs. The spacing and characteristics of the pulses shall be identical to those of 3.5.4.5.1.

*3.5.4.6 Loop Test Provisions - When loop test is enabled by a ground from the interrogator control, the synchronizer shall function normally except that SIF and Mode 4 pretriggers shall be delayed by 50 ±2.4 microseconds (50 ±5.4 µs over temperature extremes) from their normal positions and that the delay of the delayed radar trigger output to the radar shall be 462 ±3 microseconds (462 ±6.0 µs over temperature extremes) instead of 462 ±0.6 microseconds. This action shall be such that the decoded IFF replies from the associated transponder shall be displayed at a range of approximately four miles. The generation of the P_2 interrogation pulse and the ISLS control gate shall be inhibited when loop test is enabled.

3.5.4.7 Performance Monitor Function - A built-in performance monitor shall sample critical operating parameters and indicate malfunction by means of a fault indicator which shall be provided on the front panel of the synchronizer. Alarm indication shall be such as to retain the alarm signal indication in its last state when dc power is removed from the unit. When a fault exists in the synchronizer, a grounding function shall be supplied to the self-test mixer (3.5.4.7.2) where a signal shall be developed to indicate a fault in the system.

*3.5.4.7.1 Required Operational Checks - The synchronizer shall monitor the following operations, if an input radar trigger is present, once each IFF repetition period whenever the challenge function is initiated:

(1) Delayed radar trigger output presence for each radar trigger input. (Provision shall be made on the appropriate printed circuit card assembly for disabling this check when a radar trigger in excess of 2000 PRF is supplied.)

(2) Mode 4 pretrigger output presence when Mode 4 is enabled.

(3) Interrogation triggers output presence when (a) SIF is enabled, or (b) Mode 4 is enabled and Mode 4 challenge video is received.

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(4) GTC trigger output presence when (a) SIF is enabled, or (b) Mode 4 is enabled and Mode 4 GTC is received.

(5) ISLS control gate output presence, except in loop test, when (a) SIF is enabled, or (b) Mode 4 is enabled and Mode 4 ISLS gate is received.

(6) Reply video processing and bracket decoding.

3.5.4.7.2 Self-Test Mixer - Provision shall be made to mix self-test fault alarm ground signals of 0 ±1 volt from the receiver-transmitter, switch-amplifier and interrogator-computer and from within the synchronizer. (Under the no-fault condition, these signal inputs shall be either an open circuit or between +5 and +90 volts). The combined alarm ground signal of plus 3 volts or less shall be provided as an output for a 28 volt fault light indication (nominal current drain of 40 mA) at the interrogator control.

3.5.4.8 Mode and SIF Code Selection - The synchronizer shall supply dc levels of +4 ±0.4 volts and -4 ±0.4 volts to the interrogator control to be used to provide mode and SIF code enable and disable levels to the synchronizer as selected by the interrogator switches.

3.5.4.8.1 Mode Selection - The mode of operation (1, 2, 3/A, or 4) of the synchronizer shall be determined by a -4 ±0.4 volt dc level applied to the selected mode enable input from the interrogator control.

3.5.4.8.2 SIF Code Selection - For each reply pulse required to be present, as determined by the SIF code selected, the corresponding reply-pulse enable input from the interrogator control shall be at +4 ±0.4 volts dc level. When a reply-pulse is required to be missing, its corresponding enable input shall be at a -4 ±0.4 volts dc level.

3.5.4.9 Power Requirements - The synchronizer shall require +28 volts dc and 115 volts, 400 Hz single phase to operate. The total power consumption of the unit shall be 30 watts maximum.

*3.5.4.10 Dimensions and Weight - The dimensions and weight of the synchronizer shall be in accordance with Table I attached.

3.5.4.11 Controls - The following controls shall be provided on the synchronizer:

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Control	Location	Function
AC circuit breaker	Front Panel	Shall control primary ac power to the synchronizer.
DC circuit breaker	Front Panel	Shall control primary dc power to the synchronizer.

*3.5.4.12 Electrical Connections - Connections to external circuits shall be provided as follows:

Reference Designation	Receptacle Type or Approved Equivalent	Function
J1	DPJ-59C10-33P-A-2-A101 (Cannon)	Inlet and outlet for all input and output signals, respectively.
J2	UG-625B/U	Radar trigger test connector
J3	UG-625B/U	Display video test connector

*3.5.5 Receiver-Transmitter, Radio RT-988/A - This receiver-transmitter shall be identical to the RT-868A/APX-76(V) except as required to perform the following changed or added functions.

*3.5.5.1 Remote Receiver Gain Control - Provision shall be made for remote control of receiver decoding sensitivity. One control wire and one control return wire shall be provided which will allow the sensitivity to be reduced by up to 20 dB (but not in addition to the 20 dB reduction required in para. 3.4.1.2.4). It is intended that these leads be connected externally to a potentiometer (RV4SAYSD253C or equivalent). Operation of this external control shall not cause fault indications.

*3.5.5.2 Receiver Gate - The receiver shall provide a receiver gate which, when actuated by an external GTC trigger, shall return the receiver sensitivity to normal from a desensitized level of

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-12 dBm or less within 3.6 microseconds when the GTC circuitry is not enabled. The length of this gate shall be adjustable throughout the minimum range of 300 to 6200 microseconds measured from the trigger pulse. Jitter on the trailing edge of the gate shall not exceed 1% of the gate duration. At the end of the gate, the receiver sensitivity shall return to the desensitized level within a period of 75 microseconds.

*3.5.5.2.1 External Pulsed Control of Gate Duration - Provision shall be made internally selectable for control of receiver gate length by means of an external receiver gate stop pulse. When the external mode is selected, the receiver gate shall be terminated by receipt of the stop pulse. Minimum available gate length under this mode of control shall be 100 µs. Maximum available gate length shall be at least 6200 µs.

*3.5.5.2.1.1 Receiver Gate Stop Pulse - The receiver shall accept a receiver gate stop pulse which has the following characteristics when measured across 75 ohms ±10 percent.

(1)	Amplitude (on level)	+3.0 to 7.0 volts
(2)	Quiescent (off level)	0 ±0.5 volt
(3)	Polarity	Positive
(4)	Duration	1.0 to 2.0 microseconds
(5)	Impedance	75 ohms, nominal
(6)	Rise Time	0.1 microsecond, maximum
(7)	Fall Time	0.1 microsecond, maximum
(8)	Timing	Delay with respect to GTC pulse, 100 to 6200 microseconds.

*3.5.5.3 Gain Time Control - The receiver shall provide gain time control (GTC) circuitry which, when the receiver gate is triggered, shall cause the receiver sensitivity to rise from the desensitized level to an adjustable intial or GTC level that is from 20 dB to at least 42 dB less sensitive than normal sensitivity within a period of 15 microseconds measured from the receipt of the GTC trigger by the receiver gate. The receiver sensitivity shall then increase from this initial level at 15 microseconds after the receipt of the GTC trigger to within 3.0 dB of normal sensitivity in such a manner as to be within 2 dB of a 6-dB-per-octave rate-of-change curve under nominal room ambient conditions. Under extreme environmental test conditions, the receiver sensitivity shall increase in such a manner as to be within 3 dB of a 6-dB-per-octave rate-of-change curve. The first octave shall be from 15 to 30 microseconds after the receipt of the GTC trigger. The GTC circuitry shall be selectable by means of a switch on the front panel.

*3.5.5.3.1 Remote GTC On/Off Control - Provisions shall be made for remote selection of this function. The front panel switch and

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the remote switch (if used) shall be connected in parallel. GTC shall be enabled by grounding the control lead.

*3.5.5.4 Recovery Time - When the GTC gate circuit is disabled, the receiver shall recover from the effects of a transmitted signal to within 3 dB of normal sensitivity within 3.6 microseconds measured from the trailing edge of the transmitter pulse.

*3.5.5.5 RSLS Performance - The receiver-transmitter shall have an RSLS function. Two receiver input channels, the normal sum channel and the difference channel, shall be provided with the relative level of the inputs determining the presence of the normal receiver output. The normal receiver output shall be gated off unless the sum input exceeds the difference input by a ratio preset to a value settable over the range of 6 to 15 dB. This performance shall be provided over a signal input variation extending from a sum level that exceeds the specified normal sensitivity of paragraph 3.5.2.1.1.3.2 by the preset ratio plus 4 dB to a signal level of -33 dBm. The RSLS performance shall be provided at a signal frequency of 1090 MHz and under Standard Conditions, paragraph 3.3.11, with no greater than a ±4 dB change in the sum and difference channel input signal ratio permitted.

*3.5.5.5.1 Remote RSLS On/Off Control - Provision shall be made for remote selection of this function. RSLS shall be disabled by grounding of the control lead.

*3.5.5.6 Self Test Gate Output - The self-test gate shall occur at the end of the receiver gate, shall be available as an output at pin C of the front panel test connector and at a pin on J1 and have the following characteristics when terminated in 1000 ohms in parallel with 1000 pF:

(1)	Polarity	Positive
(2)	Amplitude	3.5 to 6.0 volts
(3)	Rise Time	2.5 μs maximum
(4)	Fall Time	0.5 μs maximum
(5)	Quiescent signal	0 to 0.6 volts

*3.5.5.7 Electrical Connections - Connections to external circuits shall be provided as follows:

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Reference Designation	Receptacle Type or Approved Equivalent	Function
J1	DS00-61-21P (Deutsch)	IR suppression gate, Mode 4 ISLS trigger, receiver gate stop pulse, GTC trigger, and interrogator trigger inlets, receiver video, Mode 4 reply video, Mode 4 ISLS gate, XP suppression, and self test gate outlets. Remote receiver gain control, and remote GTC on/off control.
J2	MS3113H12-10PW	Loop test enable and power inlet, Equipment fault outlet. Remote RSLS on/off control, and remote receiver gain control return.
J3	4123-0002 (Solitron)	Sum channel inlet and outlet
J4	4123-0002 (Solitron)	Difference channel inlet
J5	MS3112E16-26S	Test Connector
J6	819-B3800B-75 (Plessey)	Test Connector

*3.5.6 Base, Shock Mount, Electrical Equipment, MT-3923/APX-76(V) - Synchronizer mount, MT-3923/APX-76(V), shall provide suitable mounting for the SN-416A/APX-76(V) synchronizer to meet the vibration and shock requirements of paragraph 3.3.12 of this specification. A grounding strap shall be provided at each isolator.

*3.5.6.1 Dimensions and Weight - The maximum dimensions of the synchronizer mount shall be 4.02 x 5.77 x 9.40 inches (HxWxD) exclusive of the front retainer mechanism. The weight of the synchronizer mount shall be 1.75 lbs maximum.

*3.5.7 Base, Shock Mount, Electrical Equipment, MT-3924/APX-76(V) - Switch-amplifier mount, MT-3924/APX-76(V), shall provide suitable mounting for the SA-1568A/APX-76(V) switch-amplifier to meet the vibration and shock requirements of paragraph 3.3.12 of this specification. A grounding strap shall be provided at each isolator.

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*3.5.7.1 Dimensions and Weight - The maximum dimensions of the switch-amplifier mount shall be 2.25 x 5.77 x 11.14 inches (HxWxD) exclusive of the front retainer mechanism. The weight of the switch-amplifier mount shall be 1.75 lbs maximum.

*3.5.8 Base, Shock Mount, Electrical Equipment, MT-4024/APX-76A(V) - Receiver-transmitter mount, MT-4024/APX-76A(V), shall provide suitable mounting and forced air cooling for the receiver-transmitter to meet the vibration, shock, and cooling requirements of paragraph 3.3.12 of this specification. A grounding strap shall be provided at each of the four corner isolators.

*3.5.8.1 Dimensions and Weight - The maximum dimensions of the receiver-transmitter mount shall be 6.30 x 5.26 x 22.56 inches (HxWxD) exclusive of the front retainer mechanism. The weight of the receiver-transmitter mount shall be 5.65 lbs maximum.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements.

(1) Preproduction Tests
(2) Initial Production Tests
(3) Acceptance Tests
(4) Life Tests

4.2 Preproduction Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the approved test procedure of 4.6. The Government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptacle to the Government.

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4.2.1 Preproduction Test Data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval. On first production contracts this data shall include a list of all electrical and electronic parts giving their specified voltage, current and temperature rating and the applied circuit voltage, current and ambient and surface temperatures. The ambient and surface temperature shall be obtained under the extreme high temperature operating condition.

4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with the procedures of Specification MIL-T-5422 and interference tests in accordance with Specification MIL-I-6181.

4.2.3 Preproduction Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

4.2.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Initial Production Tests - One of the first ten production equipments shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. The preproduction sample shall not be selected for this test.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance, and quality and that it meets all applicable requirements.

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4.3.2 Accessory Material - In addition to the complete equipment submitted for Initial Production Tests the contractor shall also submit such acessory material and data necessary to test the equipment.

4.3.3 Initial Production Sample Approval - Approval of the Initial Production Sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the Initial Production Sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.4 Reconditioning of Initial Production Test Sample - On completion of the initial production test the equipment shall be reworked by the contractor by replacing all worn or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests
(2) Sampling Tests
(3) Reliability Assurance Tests
(4) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product
(2) Operational Test
(3) Manufacturing Run in Test

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4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Run in Test - Each equipment shall be operated under the conditions specified herein for a period of 12 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment. The test shall be deleted if the reliability test includes a test on each equipment which consumes at least 12 hours of operation.

Temperature	The test chamber cycle will be of a total duration of 6 hours, with a 2 hours non-operating "cold" cycle and a 4 hour operating "hot" cycle. The cycle will provide 10 minutes of equipment stabilization at the cold extreme of -54°C and two hours of stabilized operation at the hot extreme of +71°C.
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 60 Hz (excluding resonant points) and a minimum amplitude of ± 3 g's.

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes during each hour of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane. During the 6-hour period of operation following a 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A

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record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.4.2 Sampling Tests - Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the government inspector in accordance with the following: (Sampling tests shall be conducted only when Reliability Assurance Tests of 4.4.3 are deleted by contractual action.)

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Sampling Test
First 10	0*
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof.

*NOTE: If by contract action the Initial Production Test (4.3) is deleted, then a Sample Test shall be conducted on one equipment from the first 10 produced.

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, initial production, individual and special tests.

(3) Manufacturing run in test specified in paragraph 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

4.4.3 Reliability Assurance Tests - Reliability Assurance Tests shall be conducted using MIL-STD-781. Tests as required by both the Qualification Phase and the Sampling Phase shall be conducted. Classification of failure shall be in accordance with MIL-STD-781 and AR-34.

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*4.4.3.1 Qualification Phase - Prior to the acceptance of equipments under the contract or order, a minimum of three (3) equipments shall be tested as outlined in MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests". The maximum number of equipments to be used shall be those listed in Table 5 of MIL-STD-781. For the Qualification Phase, Test Level F shall be used. The Accept-Reject Criteria for Test Plan III shall be used.

*4.4.3.2 Reliability Production Acceptance (Sampling) Phase Tests - The equipment, throughout production, shall be tested as outlined in MIL-STD-781 (as modified herein) under the section entitled "Production Acceptance (Sampling) Phase of Production Reliability Tests". Test level F shall be used.

*4.4.3.2.1 All Equipment Test - An All Equipment Screening Test shall be performed in accordance with MIL-STD-781, Test Plan XXIX with the following details.

*4.4.3.2.1.1 Test Duration - Each equipment produced, except those submitted for the Reliability Qualification Test, shall be tested for 72 hrs.

*4.4.3.2.1.2 Burn-In - If a burn-in period is to be used, the details thereof shall be included in the approved test procedure.

*4.4.3.2.1.3 Odd Quantity Items - When odd quantity items (i.e. less than complete equipments) are produced, each item shall be tested for 72 hours either as part of a complete equipment or as a separate item, at the option of the contractor. If an odd quantity item is tested as part of a complete equipment, the complete equipment's operating test hours and failures shall be totaled and the results included within the test evaluation data required by Test Plan XXIX (in this case certain items will be tested for more than one period of 72 hours). If odd quantity items are tested separately, the items' operating test hours and failures shall be added to the total operating test hours and total number of failures for all identical items tested as part of complete equipments and the net results evaluated by the criteria of Test Plan XXIX using the appropriate individual item MTBF of paragraph 3.3.3.3.1 as the specified MTBF.

*4.4.3.2.1.4 Test Reporting - At the conclusion of each month, the test results shall be sent to the procuring activity and to the Naval Air Systems Command, Attention: Avionics Division.

*4.4.3.2.1.5 Reject Situation - The procuring activity reserves the right to stop the acceptance of equipment at any time that an equipment reject situation exists pending a review of the contractor's efforts to improve the equipment, the equipment parts, the equipment workmanship, etc., so that the entire compilation will show other than a reject decision.

4.4.3.3 Test Details - The test details, including the following, shall be part of the test procedures to be submitted and approved by the procuring activity prior to the beginning of the Qualification Test Phase of the Reliability Assurance Tests.

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4.4.3.3.1 Test Level - Test level F specified in MIL-STD-781 shall apply, except that the interrogator control may be subject to prevailing ambient conditions near the test chamber. The test chamber cycle will be a total duration of 6 hours, with a 2 hour "cold" cycle and a 4 hour "hot" cycle. The cycle will provide 10 minutes of equipment stabilization at the cold extreme of -54°C and two hours of stabilized operation at the hot extreme of +71°C.

4.4.3.3.2 Performance Characteristics to be Measured - The performance characteristics selected from 3.4 and specified in the contractor's approved reliability test plan shall be measured on a once-per-working day basis. Inability of the equipment to meet approved test criteria for the characteristics under test shall be considered a failure.

4.4.3.3.3 Preventive Maintenance - During the period of the tests, preventive maintenance measures may be performed upon the equipment as determined by the design activity with the approval of the procuring agency.

4.4.3.3.4 Equipment Failure - Should a failure occur during either the sampling, or reliability tests, the following action shall be taken:

(1) Determine the cause of failure

(2) Determine if the failure is an isolated case or design defect. In the case of a design defect:

(a) Submit to the procuring agency for approval proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(b) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.4.4 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

4.4.4.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate additional tests are required. (This will be determined by the procuring activity.)

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4.4.4.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.5 Equipment Failure - Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 300-hours duration and shall be conducted on equipments that have passed the individual test. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the government inspector in accordance with the following. (Equipments which have successfully passed the Initial Production Test, Sampling Tests, Reliability Tests, or Special Tests may be selected for life tests.) When reliability tests are conducted, the life test shall be omitted if, during the reliability tests, a quantity of equipments equal to, or more than, that listed below receive at least 300 hours each of test time.

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof.

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions:

Temperature	Normal Room
Altitude	Normal ground (0-5000 ft)

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AC Voltage	115 ±5 volts (at lowest applicable frequency)
DC Voltage	27.5 ±2.0 volts

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep the daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part type number

(2) The circuit reference symbol number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure.

4.5.4.1 Failure Report - In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

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4.6 Test Procedures - The procedures used for conducting reproduction tests, acceptance tests and life tests shall be repared by the contractor and submitted to the procuring activity or review and approval. The right is reserved by the procuring ctivity or the Government inspector to modify the tests or require ny additional tests deemed necessary to determine compliance with he requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test rocedures. When approved test procedures are available from revious contracts such prodcedures will be provided and may be sed when their use is approved by the procuring activity. However, he right is reserved by the procuring activity to require modification of such procedures, including additional tests, when eemed necessary.

4.7 Reconditioning of Tested Equipment - Equipment which has een subjected to acceptance and life tests shall be reconditioned y the contractor by replacing all wear or damaged items. After eworking the contractor shall resubmit the equipment for acceptance.

4.8 Presubmission Testing - No item, part or complete equipent shall be submitted by the contractor until it has been previusly tested and inspected by the contractor and found to comply, o the best of his knowledge and belief, with all applicable equirements.

4.9 Rejection and Retest - Equipment which has been rejected ay be reworked or have parts replaced to correct the defects and esubmitted for acceptance. Before resubmitting, full particulars oncerning previous rejection and the action taken to correct the efects found in the original shall be furnished the Government nspector.

*4.10 Quality Assurance Provisions for Control, Interrogator et, C-7994/APX-76A(V) - The Quality Assurance Provisions shall e in accordance with SECTION 4 of this specification and paraaph 4 of Exhibit SENX-LM-66-1, dated 5 August 1966. When the equirements conflict, the lighting requirements of Exhibit SENX-M-66-1 shall govern.

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5. PREPARATION FOR DELIVERY

*5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with specifications MIL-E-17555 and MIL-STD-794. In the event the equipment is not covered in Specification MIL-E-17555, the method of preservation for Level A shall be determined in accordance with the selection chart in Appendix D of MIL-STD-794.

6. NOTES

6.1 Intended Use - The equipment shall provide Air-by-Air radar target identification capabilities for use in all military aircraft to carry out specific military missions.

*6.2 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(1) Title, number, and date of this specification.

(2) Selection of applicable levels of packaging and packing. (See 5.1)

6.3 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

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(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein, all referenced specifications shall apply to the extent specified.

6.4 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specifications are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete support information.

6.5 Non-Repairable Subassemblies - As a general rule non-repairable subassemblies should be encapsulated or hermetically-sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the non-repairable subassembly should evidence a Mean-Time-to-Failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

*6.6 Type Designations - The type designation may be modified by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.3.10. The correct type number shall be used on nameplates, shipping records and instruction books, as applicable.

6.7 Asterisk - In specification revisions and superseding amendments an asterisk "*" preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue.

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This has been done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the asterisk notations and relationship to the last previous issue.

6.8 Definitions - Unless further amended by the individual specification paragraph, these definitions shall apply.

6.8.1 Interrogation - An interrogation is two (2) or more radio frequency pulses transmitted by an interrogator set.

6.8.2 Reply - A reply is a radio frequency pulse or pulse group transmitted by a transponder set as a result of an interrogation.

6.8.3 Pulse Characteristics

6.8.3.1 Pulse Amplitude - Pulse amplitude is the maximum voltage amplitude of the pulse or the rf pulse envelope, excluding spike.

6.8.3.2 Pulse Amplitude Jitter - Pulse amplitude jitter is the variation of the pulse amplitude of successive pulses caused by unintentional effects within the equipment expressed in percent of the average of the pulse.

6.8.3.3 Pulse Duration (or Width) - Pulse duration is that interval of time expressed in microseconds, between the points at which the amplitude of the pulse is 50 percent of the pulse amplitude.

6.8.3.4 Pulse Rise Time - Pulse rise time is that interval of time, expressed in microseconds, required for the leading edge of the pulse to rise from 10 to 90% of the pulse amplitude.

6.8.3.5 Pulse Decay Time - Pulse decay time is that interval of time, expressed in microseconds, required for the trailing edge of the pulse to decay from 90 to 10% of the pulse amplitude.

6.8.3.6 Pulse Spacing - Pulse spacing is that interval of time, expressed in microseconds, between 50% pulse amplitude points on the leading-edge of the pulses involved.

6.8.4 Signal Strength - Signal strength is specified in dB referred to 1 mW.

6.8.5 Squitter - Squitter is decoding of noise or similar interference.

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6.8.6 Railing - Railing is high duty cycle pulse interference not correctly coded.

6.8.7 Hot Switching - Switching while conducting full rf power.

*6.9 Dimensions and Weight of Interrogator Set - See Table I

*6.10 This Specification is under the cognizance of AIR 53341C.

6.11 Associated Equipment

*6.11.1 Interrogator Computer, KIR-1A/TSEC - The interrogator computer is the KIR-1A/TSEC computer. Input/Output characteristics of this component comply with DOD-AIMS 64-900A.

6.11.2 Interrogator Antenna - An interrogator antenna integrated with the radar antenna by modification kit shall normally be used by the interrogator set. The design of the antenna shall be adequate to withstand the radar antenna environment requirements.

6.11.2.1 Radar Performance Degradation - The inclusion of the interrogator antenna shall not degrade the radar antenna performance by more than the following nominal amounts:

(1)	Gain reduction	0.5 dB
(2)	Boresight shift	Negligible: Consistent with the radar requirements.
(3)	Grating lobe level	Down 27 dB
(4)	Reflection coefficient	0.05

6.11.2.2 Interrogator Antenna L-Band Characteristics - The antenna shall have a sum and difference mode in azimuth and the following characteristics (nominal for a 32-inch aperture antenna):

(1)	Polarization	vertical
(2)	Gain (sum mode)	12 dBI at pattern peak
(3)	Gain (difference mode)	3 dB below sum mode
(4)	Beamwidth - 3 dB (sum mode)	(a) elevation, 35° (b) azimuth, 25°
(5)	Difference/sum pattern crossover	28° beamwidth

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(6)	Effective beamwidth (sum pattern 9 dB above difference pattern null)	13°
(7)	Difference pattern coverage	the difference pattern level shall be greater than the sum pattern level at all points outside crossover where sum pattern is less than 20 dB down
(8)	Frequency	1030 ±10 MHz 1090 ±10 MHz
(9)	VSWR	2 to 1 maximum
(10)	Isolation	20 dB minimum
(11)	Peak power	(a) sum mode +33 dBW (b) difference mode +38 dBW

6.11.3 Radar Modification Kit - The radar shall be made compatible with the interrogator set trigger, control and display video requirements as described in sections 3.4.2.2, 3.4.2.3, 3.4.2.4, 3.4.3.1, 3.4.3.4, and 3.4.3.6 by means of a radar modification kit.

6.11.3.1 Radar Trigger Bypass - Provisions shall be incorporated in the design of the radar modification kit such that with the IFF system in the off condition the radar trigger circuit is returned to its unmodified state.

6.11.4 Radome Modification Kit - The radome of the aircraft shall be made compatible with the interrogator set transmissions and receptions with negligible degradation of radar operation by means of a radome modification kit.

*6.12 Receiver-Transmitter RT-901/APX-83(V) - The requirements for this Receiver-Transmitter are identical to those of the RT-988/A, defined in paragraph 3.5.5, except as specified by additional U.S. Air Force requirements. See DOD AIMS 69-253.

Custodian:
NAVY-AS

Preparing Activity
NAVY-AS

Project No. 5895-N262

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TABLE I - DIMENSIONS AND WEIGHT OF INTERROGATOR SET EQUIPMENT GROUP COMPONENTS

Component	H X W X D (Note) Inches	Volume Cu. In.	Weight Lbs.
(1) Control, Interrogator Set	3 X 5-3/4 X 3-1/4 (1,2)	55	1-3/4
(2) Receiver-Transmitter, Radio	7-5/8 X 5 X 17-1/2 (1,2)	670	19
(3) Switch-Amplifier	6 X 5-1/8 X 9-7/8 (1)	305	9
(4) Synchronizer, Electrical	6 X 5-1/8 X 7-1/2 (1,2)	230	7-3/8
(5) Interrogator Computer KIR-1A/TSEC (for reference only)	7-5/8 X 4-7/8 X 7-1/4	270	20

NOTE: (1) Exclusive of protrusions in front of panel, and guide rails on unit sides.

(2) Exclusive of connector or air inlet protrusion at rear.

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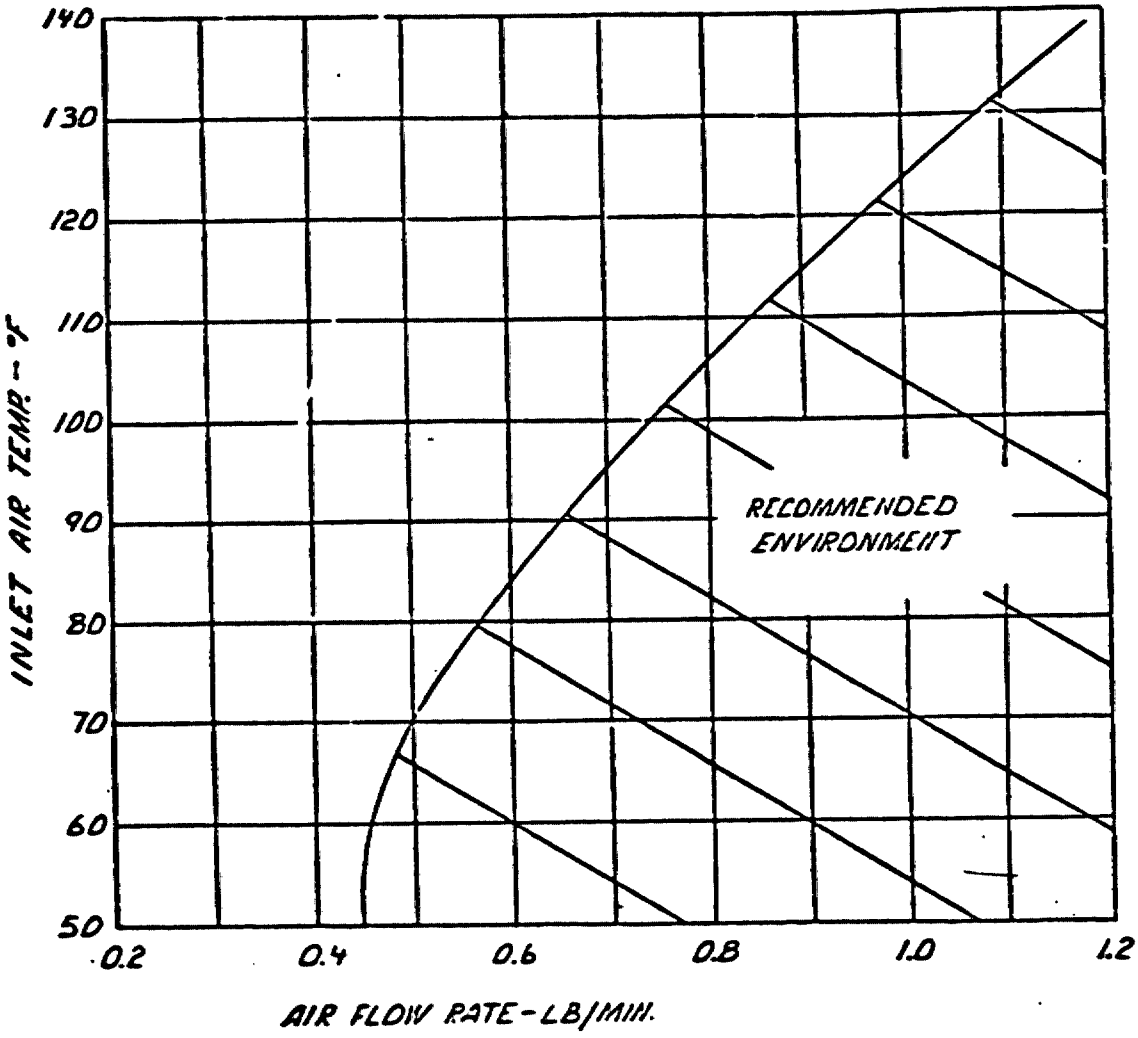


Figure 1. Inlet Temperature vs Air Requirements for Receiver-Transmitter, Transmitting at 1.0% RF Duty Cycle

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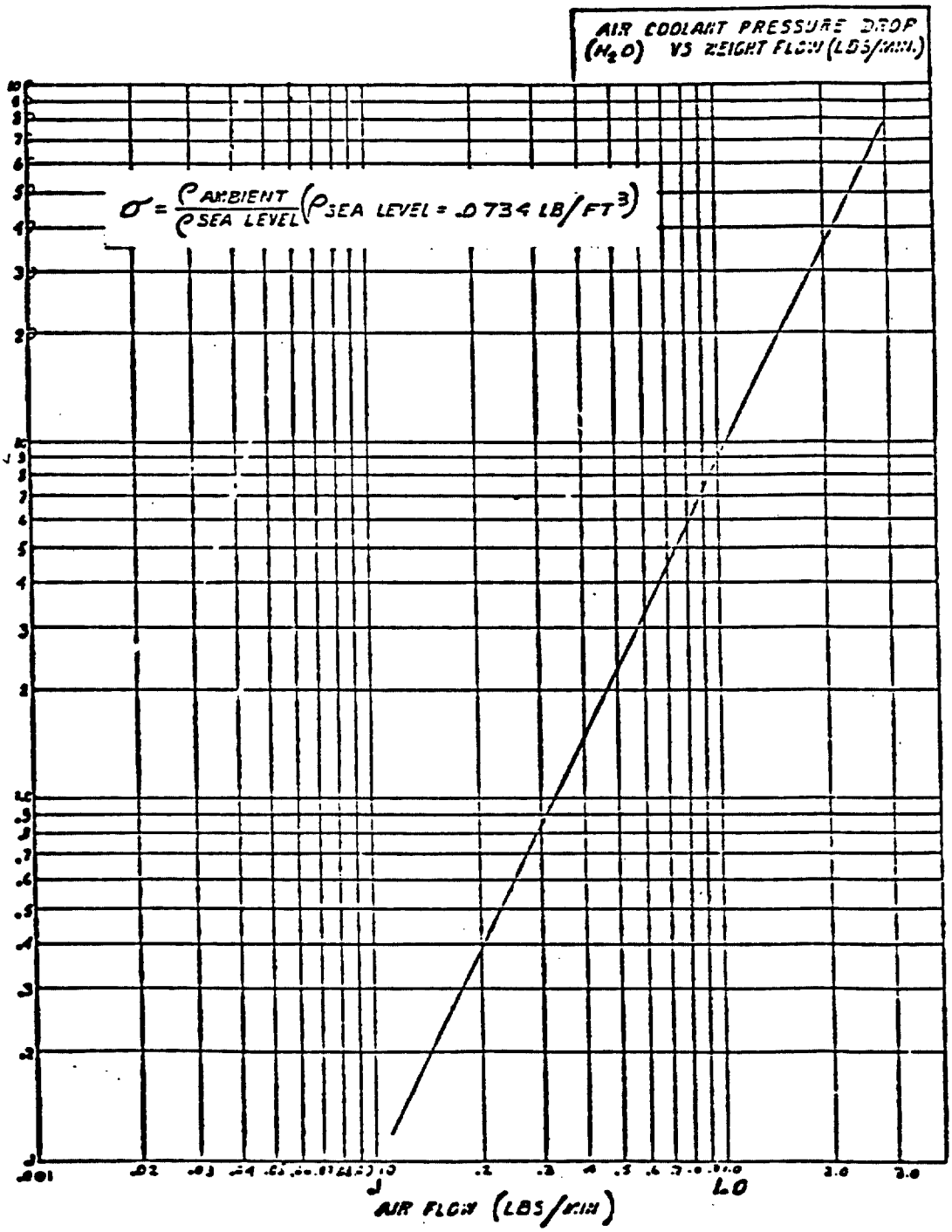


Figure 2. Air Coolant Pressure Drop vs Weight Flow, Receiver-Transmitter

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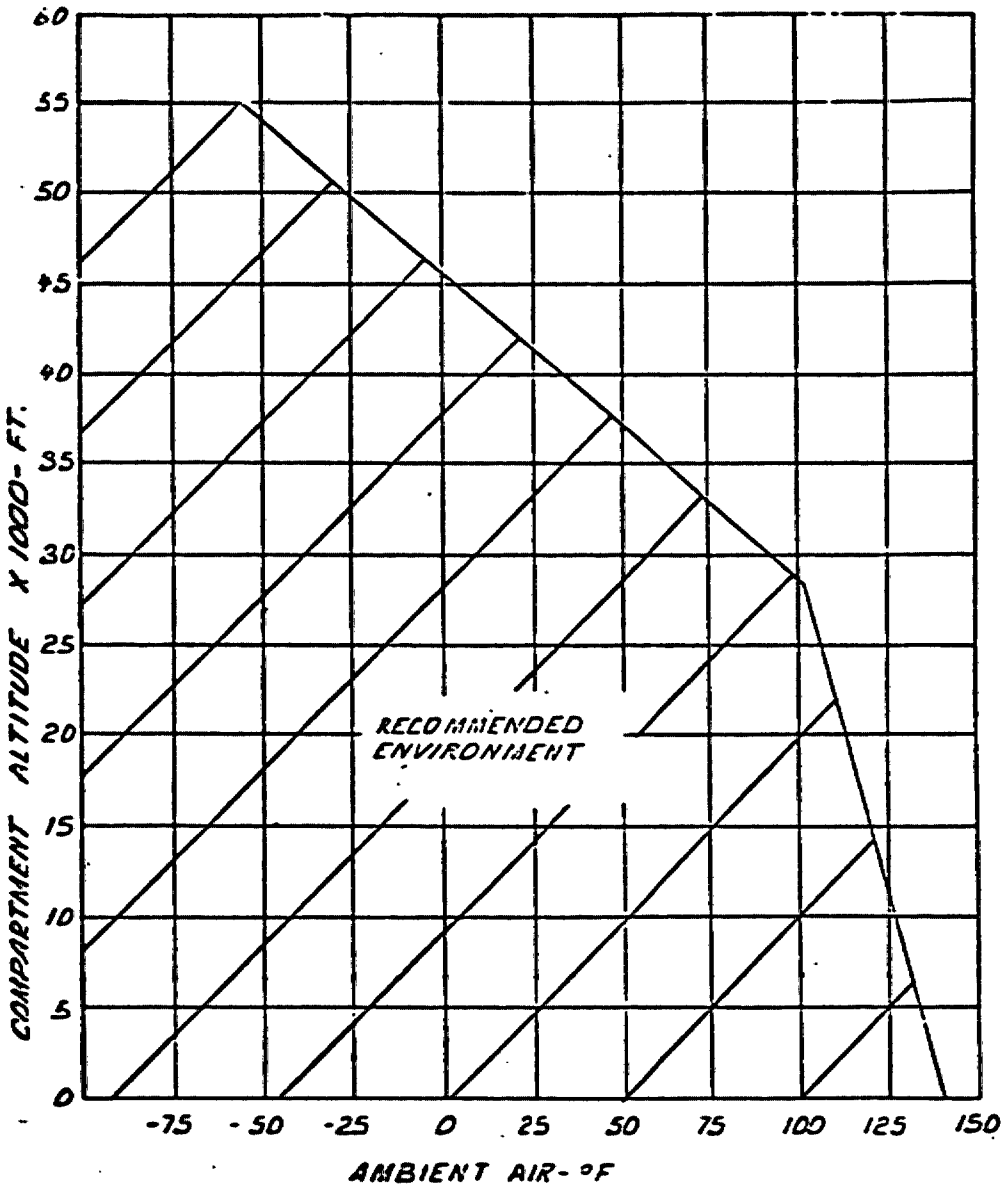


Figure 3. Altitude vs Air Requirements for Receiver-Transmitter on Mount MT-4024/APX-76A(V), Transmitting at 0.6% RF Duty Cycle

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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DOCUMENT IDENTIFIER AND TITLE MIL-I-81453A(AS) Interrogator Set AN/APX-76A(V)

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	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

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   A. GIVE PARAGRAPH NUMBER AND WORDING.

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